NIBB ゲノムインフォマティックス・トレーニングコース2025 夏
「RNA-seq入門」
2025.9.10-2025.9.11

NGS可視化ツールIGV

基礎生物学研究所
超階層生物学センター
トランスオミクス解析室
山口勝司

データ可視化ツール・IGVの紹介・実習

IGV
Integrative Genomics Viewer

IGV
IGV desktop application

IGV Web App
IGV in a web browser

Juicebox Web
Hi-C contact map viewer

https://igv.org

For Developers

igv.js	Use igv.js to embed an interactive genome visualization component in your web app.
igv-notebook	A Python package that wraps igv.js for embedding in an IPython notebook. Supports both Jupyter and Google Colab.
igv-reports	Generate self-contained HTML reports that consist of a table of genomic sites and associated IGV views for each site.

GitHub repos: IGV Desktop | IGV Web App | igv.js | igv-notebook | igv-reports | Juicebox Web | juicebox.js

Citing IGV

To cite your use of IGV in your publication, please reference one or more of:

James T. Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S. Lander, Gad Getz, Jill P. Mesirov. Integrative Genomics Viewer. Nature Biotechnology 29, 24–26 (2011). A public access version is also available: PMC3346182.

Helga Thorvaldsdóttir, James T. Robinson, Jill P. Mesirov. Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration. Briefings in Bioinformatics 14, 178-192 (2013).

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Contact Us

The IGV Team is based at UC San Diego and the Broad Institute of MIT and Harvard. The best way to reach us for support questions, bug reports, feature requests, and suggestions is by posting to the igv-help forum or by creating new issues in our GitHub repositories. To reach us privately on other topics not related to general support, feel free to email us directly.

Funding

IGV development has been supported by funding from the National Cancer Institute (NCI) of the National Institutes of Health, the Informatics Technology for Cancer Reserach (ITCR) of the NCI, and the Starr Cancer Consortium.

IGV desktop

IGV

IGV desktop application

IGV Desktop Application

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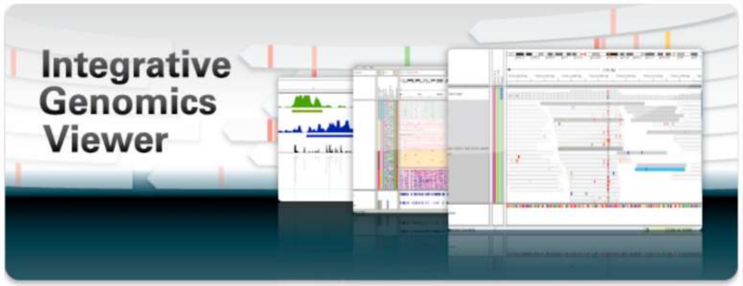

https://igv.org/doc/desktop

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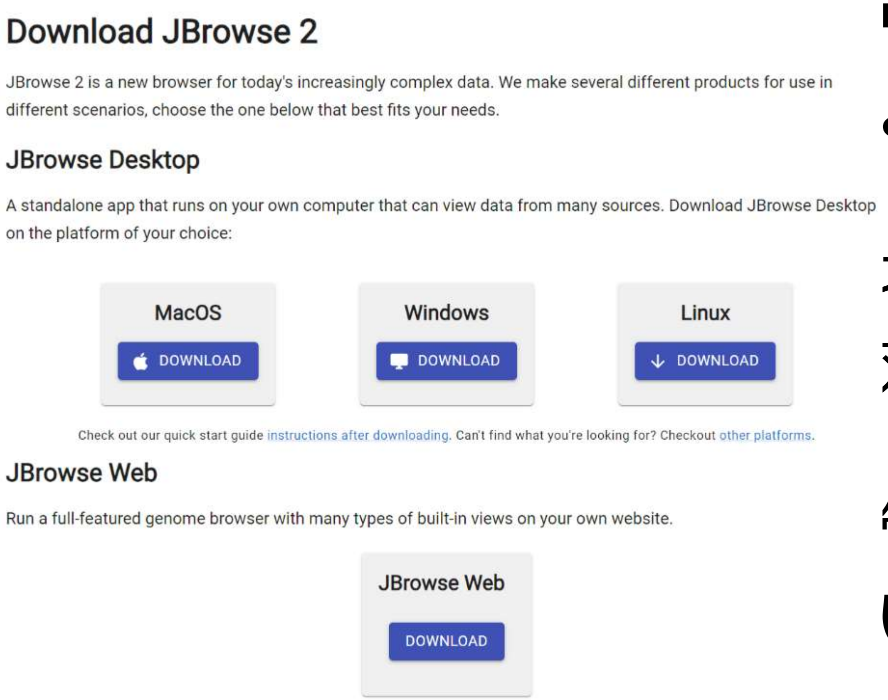
なぜIGVを取り上げるか

データ可視化ツール　2つのタイプ

・自分のパソコン(ローカル環境)にインストールして使うタイプ

・サーバーに構築して、ネットワーク上で使うタイプ

後者はコミュニティーでの利用目的、ウェブ公開を目的とするには良い。より大容量なデータに対応できる。



ネットワーク・情報セキュリティに対する高度な知識が要求される。

管理者的な人がいるなら、これも良いが・・・

JBrowseも現在はIGV的な利用も可

JBrowse desktop

もっとお手軽なものとしてIGVを紹介

可視化ツールに求められるものは何か

膨大なデータを、如何に直感的、統合的に理解できるか
sortや絞り込みで定量的に理解する表データと対比双璧になる

多様なデジタル情報
・配列、GC ratio、遺伝子情報
・遺伝子発現情報
・遺伝子多型の位置情報・頻度情報
・様々なデータの精度情報
・RNA-seq, ChIP-seq, RAD-seq, BS-seq・・・

レファレンス配列 / gene model / gene annotationと
NGSデータを並べて比較
複数のデータセットを並べて比較

様々なスケールで比較・統合的に解釈できるようにしたい

ゲノムviewerに自分のデータを乗せ、
統合的直感的に比較・解釈できること

可視化ツールをどう選ぶか

選択の基準
genome data viewing に求められるもの
取捨選択の基準

1. 無料 / 有料 / 基本無料

2. 個人的レベルの使用 / コミュニティーレベルの使用

3. 見るだけ/自分から色々工夫

4. アクセスビリティ・ユーザビリティ
   - 導入に必要なコンピュータスペック
   - マニュアルは分かりやすいか
   - 情報の多さ
   - 利用の簡便さ
   - 使っている人が近くにいるか

Integrative Genomics Viewer(IGV)

お手軽ツール

・無料
・多くの人が使っている、情報も多い
・javaのプログラムなので、オールプラットフォーム対応
・マニュアルは親切、サンプルデータもある
・WEBサーバーではなく、PCレベルで利用出来る
・データ閲覧環境の共有も可能

誰もが簡便に使えるものが良い。

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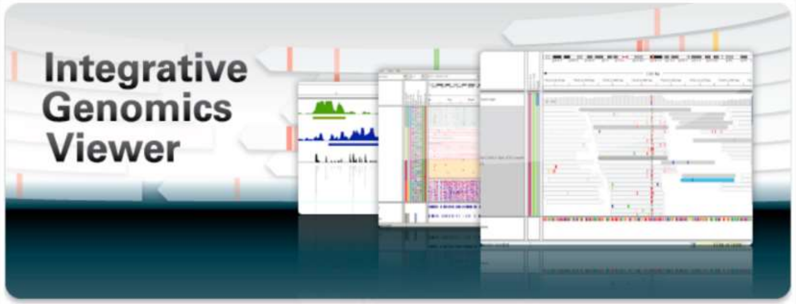

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NATURE BIOTECHNOLOGY | OPINION AND COMMENT | CORRESPONDENCE

Integrative genomics viewer

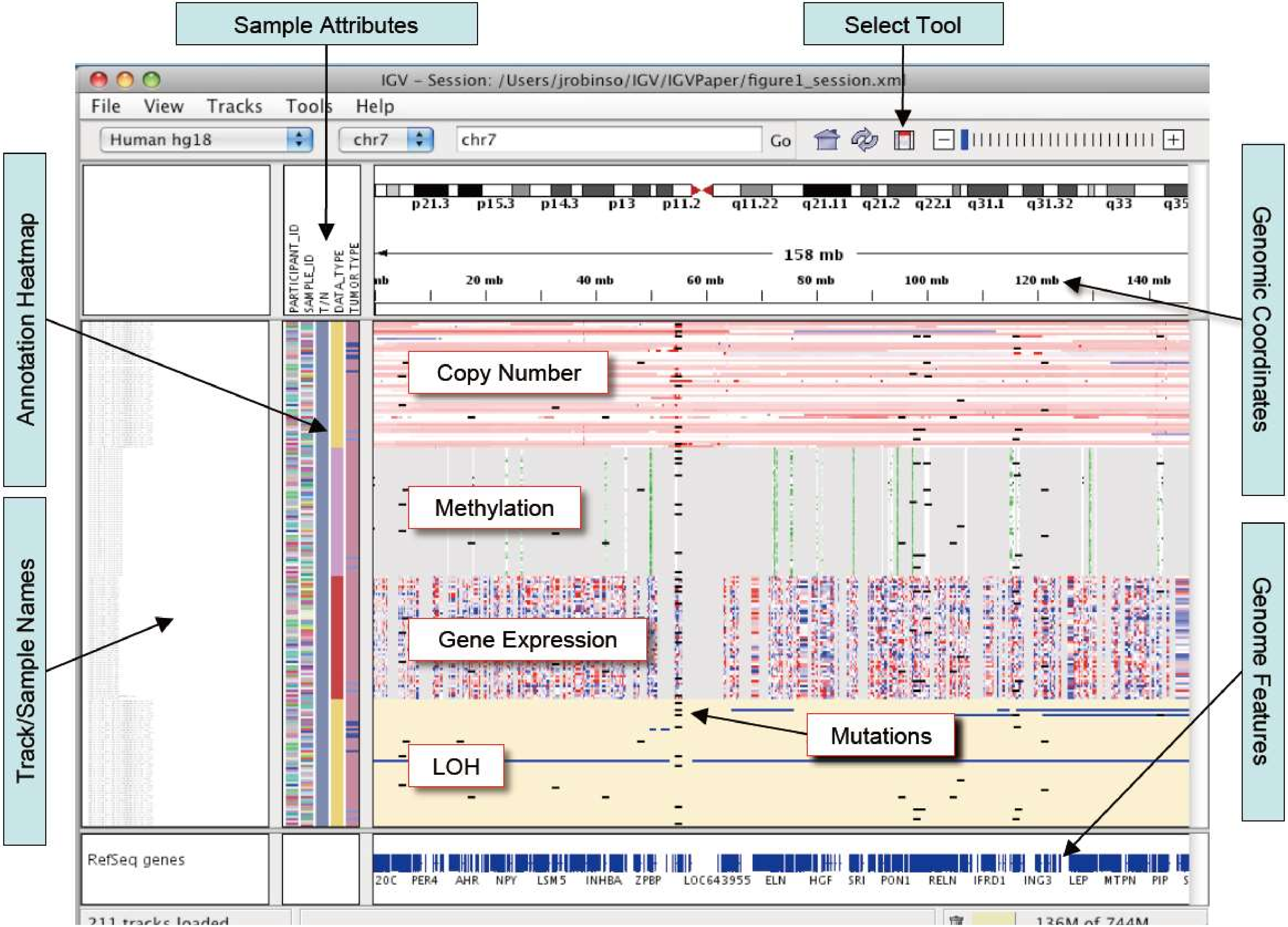
James T Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S Lander, Gad Getz & Jill P Mesirov

Affiliations | Corresponding authors

Nature Biotechnology **29**, 24–26 (2011) | doi:10.1038/nbt.1754
Published online 10 January 2011

To the Editor:

Rapid improvements in sequencing and array-based platforms are resulting in a flood of diverse genome-wide data, including data from exome and whole-genome sequencing, epigenetic surveys, expression profiling of coding and noncoding RNAs, single nucleotide polymorphism (SNP) and copy number profiling, and functional assays. Analysis of these large, diverse data sets holds the promise of a more comprehensive understanding of the genome and its relation to human disease. Experienced and knowledgeable human review is an essential component of this process, complementing computational approaches. This calls for efficient and intuitive visualization tools able to scale to very large data sets and to flexibly integrate multiple data types, including clinical data. However, the sheer volume and scope of data pose a significant challenge to the development of such tools.



Nature Biotech. 29:24–26 (2011) Supplement figureからの抜粋

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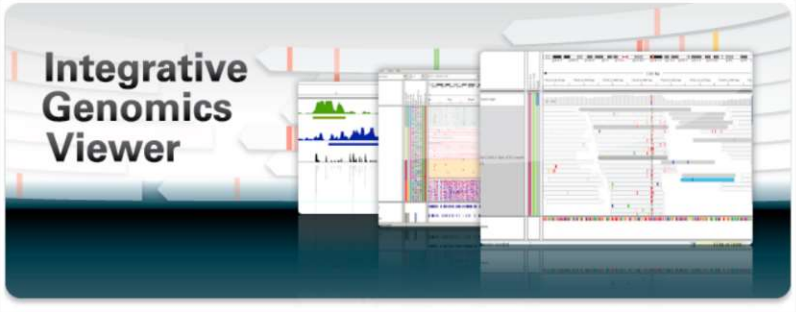

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    - BED
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    - BEDPE
    - bigBed
    - bigGenePred
    - bigNarrowPeak
    - bigWig
    - Birdsuite Files
    - broadPeak
    - CBS
    - Chemical Reactivity Probing Profiles ◂
    - CN
    - CRAM
    - genePred
    - GFF/GTF
    - GISTIC
    - GWAS
    - IGV
    - LOH
    - MAF (Multiple Alignment Format)
    - MAF (Mutation Annotation Format)
    - Merged BAM File
    - narrowPeak
    - PSL
    - RNA Secondary Structure Formats ◂
    - SAM
    - SEG
    - TDF
    - Track Lines
    - Type Lines
    - VCF
    - WIG

Download IGV

IGV version 2.19.5

IGV for Windows
Java included

What's New: See the Release Notes for what's new in each IGV release.

All platforms

IGV requires Java 21 or greater as of IGV 2.19.1. If you download one of the IGV versions that does not include Java, make sure you have Java 21 installed and in your path.

Mac users: As of IGV 2.19.1, the IGV Mac apps require **MacOS 11 (Big Sur)** or greater.

- To run **the latest IGV** on older MacOS versions, download and use the ***Command line IGV for all platforms*** option from the main Download page.
- To run **a working IGV Mac app** on older MacOS versions, download the **2.18.4 Mac app** (includes Java). Or load any IGV Mac app version 2.18.4 or earlier from the IGV download archive.

Linux users: The *IGV for Linux* download includes AdoptOpenJDK (now Eclipse Temurin) version 21 for x64 Linux. See their list of supported platforms here. If your platform is not on the "x64 Linux" list, or the packaged Java does not work on your version of Linux, download the *"Command line IGV for all platforms"* and use it with your own Java installation.

IGV for MacOS
Apple Chip - Java included

IGV for MacOS
Intel Chip - Java included

IGV for MacOS
Separate Java 21 required

IGV for Windows
Java included

IGV for Windows
Separate Java 21 required

IGV for Linux
Java included

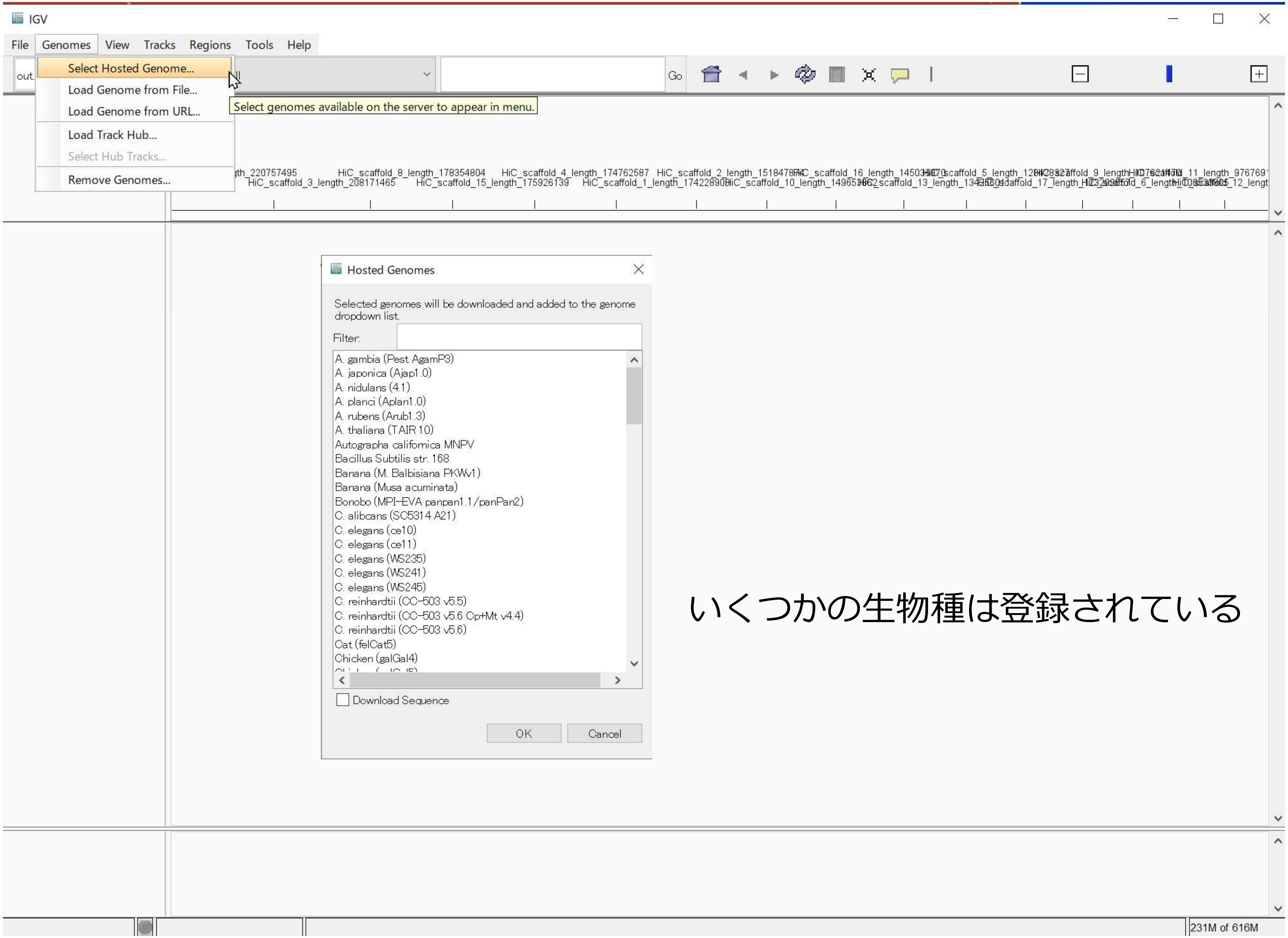
Command line IGV and igvtools for all platforms
Separate Java 21 required

利用するパソコンの
プラットフォームに応じて選択

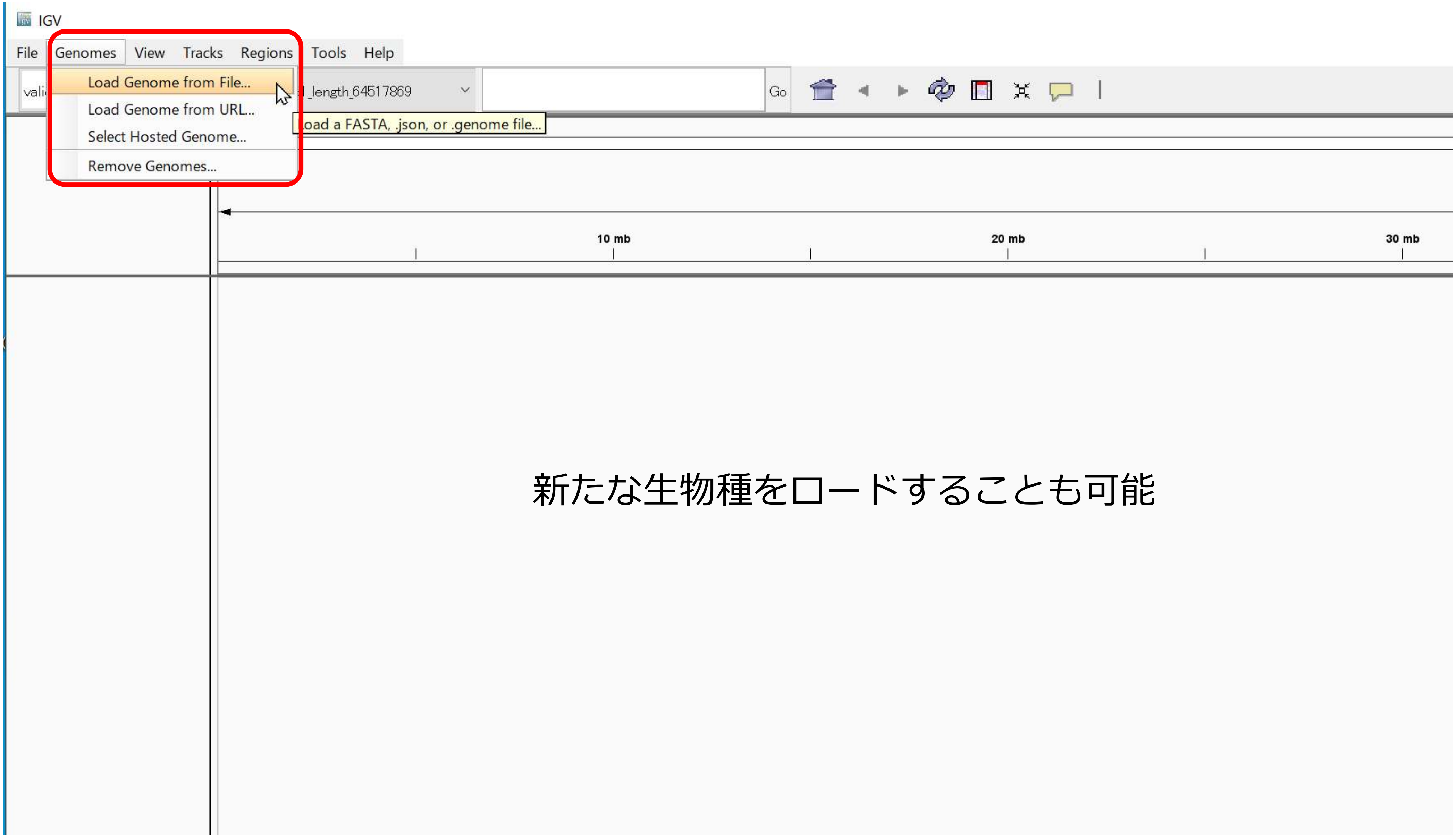
Other releases of IGV Desktop

Development snapshot build. Latest development snapshot; built at least nightly

Archived releases. Old releases going back to IGV 2.0



いくつかの生物種は登録されている



新たな生物種をロードすることも可能

ゲノムViewerなので次世代ＤＮＡシーケンサーのデータに限定されない。
マイクロアレイの結果や、ゲノムアノテーションの情報も随時表示できる。

対応するファイル形式に応じて、表示方法が決まる。

File Formats

- BAM
- BED
- BEDPE
- BedGraph
- bigBed
- bigWig
- Birdsuite Files
- broadPeak
- CBS
- Chemical Reactivity Probing Profiles
- chrom.sizes
- CN
- Custom File Formats
- Cytoband
- FASTA
- GCT
- CRAM
- genePred
- GFF/GTF
- GISTIC
- Goby
- GWAS
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- MUT
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- PSL
- RES
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- SAM
- Sample Info (Attributes) file
- SEG
- TDF
- Track Line
- Type Line
- VCF
- WIG

File Formats

IGV supports a number of different file formats for experimental data and genome annotations. For a complete list of supported formats see http://www.broadinstitute.org/igv/FileFormats. The following table shows the recommended file formats for a number of common data types.

Source Data	Recommended File Formats
ChIP-Seq, RNA-Seq	WIG, TDF
Copy number	CN, SNP, TDF, canary_calls (Birdsuite)
Gene expression data	GCT, RES, TDF
Genome annotations	GFF, BED, GTF, PSL, UCSC table format
GISTIC data	GISTIC
LOH data	LOH, TDF
Mutation data	MUT, MAF
Variant calls	VCF
RNAi data	GCT
Segmented data	SEG, CBS
Sequence alignment data	BAM, SAM, PSL
Any numeric data	IGV, WIG, TDF
Sample metatadata	Tab-delimited sample info file

その他の便利機能

セッションの保存
- 表示しているデータの読み込み状況を、それごと保存。
- セッションをロードすることで、意図した画面を表示できる。
  - データセットが揃っていること、フォルダー構造が同一である必要がある。

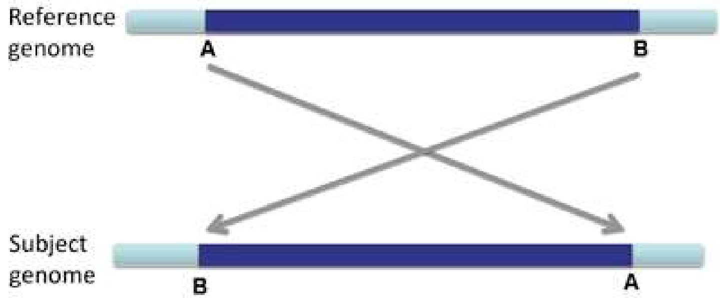
Ref配列のコピー
- Define a region of interestのアイコンを押す。
- カーソルが十字となる。
- 配列の取り出したい領域の開始と終了ポジションをクリック。
- 指定された領域が赤色になり、右クリックしてCopy Sequence

バッチ処理
- 重要領域の画面スナップショットを自動で取ったりできる。

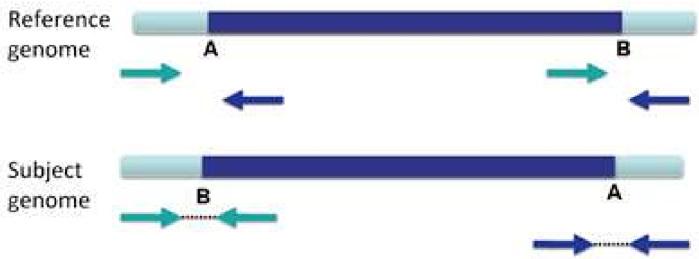
```
new
load myfile.bam
snapshotDirectory mySnapshotDirectory
genome hg18
goto chr1:65,289,335-65,309,335
sort position
collapse
snapshot
goto chr1:113,144,120-113,164,120
sort base
collapse
snapshot
```

Inversions

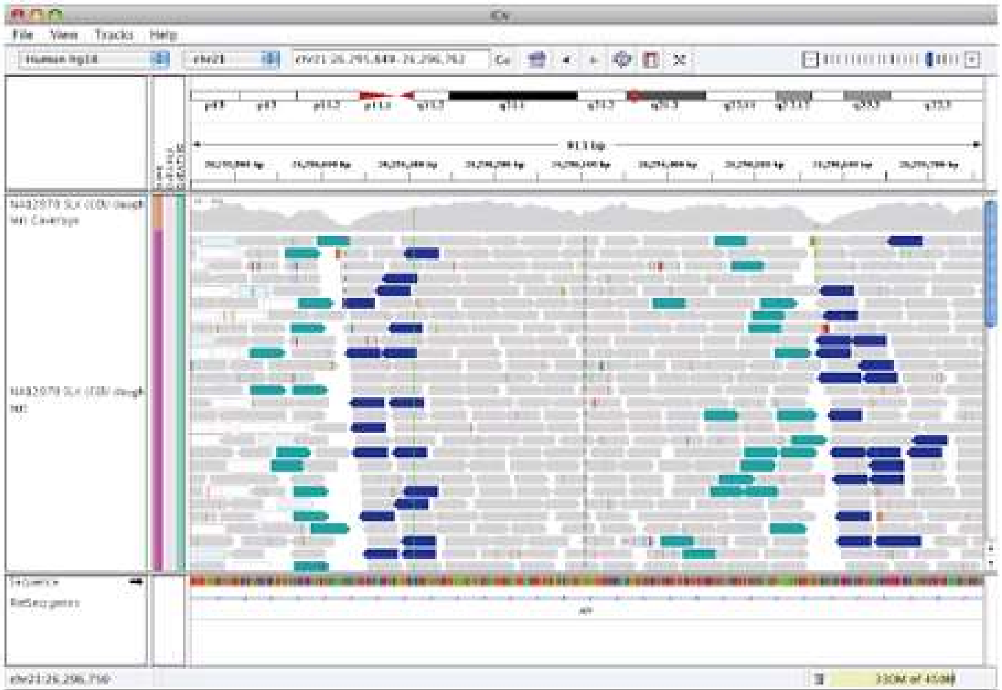
An inversion is a large section of DNA that is reversed in the subject genome compared to the reference genome.



When an inversion shows up in paired-end reads, the reads are distinctively variant from the reference genome.



This appears in IGV as shown below.



Interpreting Color by Insert Size

The inferred insert size can be used to detect structural variants, such as:

- deletions
- insertions
- inter-chromosomal rearrangements

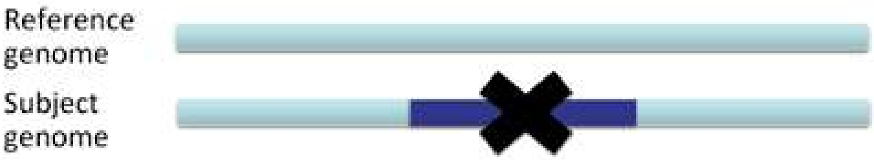
IGV uses color coding to flag anomalous insert sizes. When you select Color alignments>by insert size in the popup menu, the default coloring scheme is:

-  for an insert that is larger than expected
-  for an insert that is smaller than expected
- 

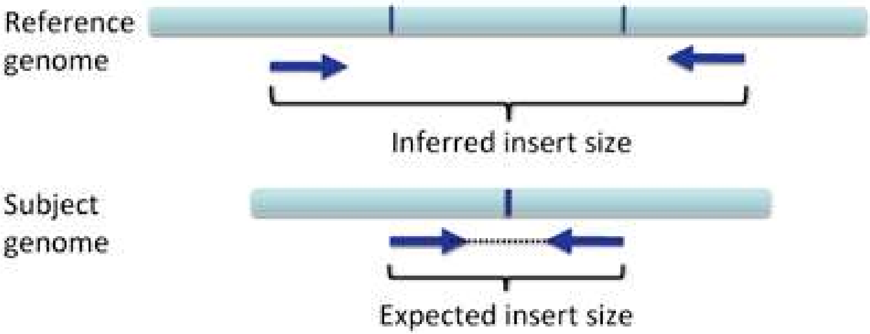
for paired end reads that are coded by the chromosome on which their mates can be found

Deletions

A deletion is a large section of DNA that is absent in the subject genome compared to the reference genome.



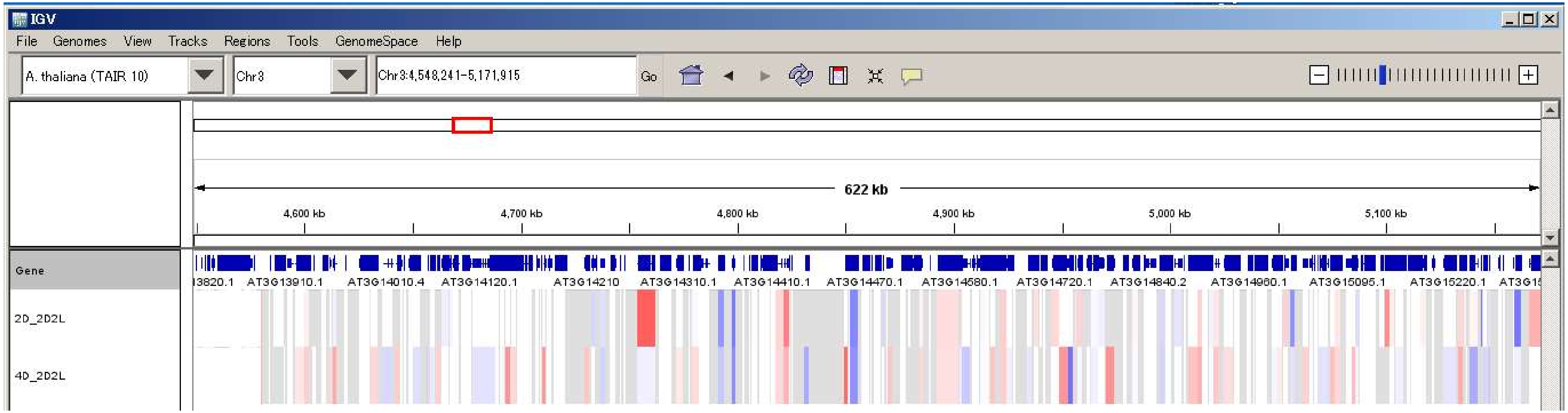
The "expected" insert size is the insert size obtained in sequencing the subject genome. The "inferred" insert size is the insert size that would result in the reference genome, assuming the same pair of reads.



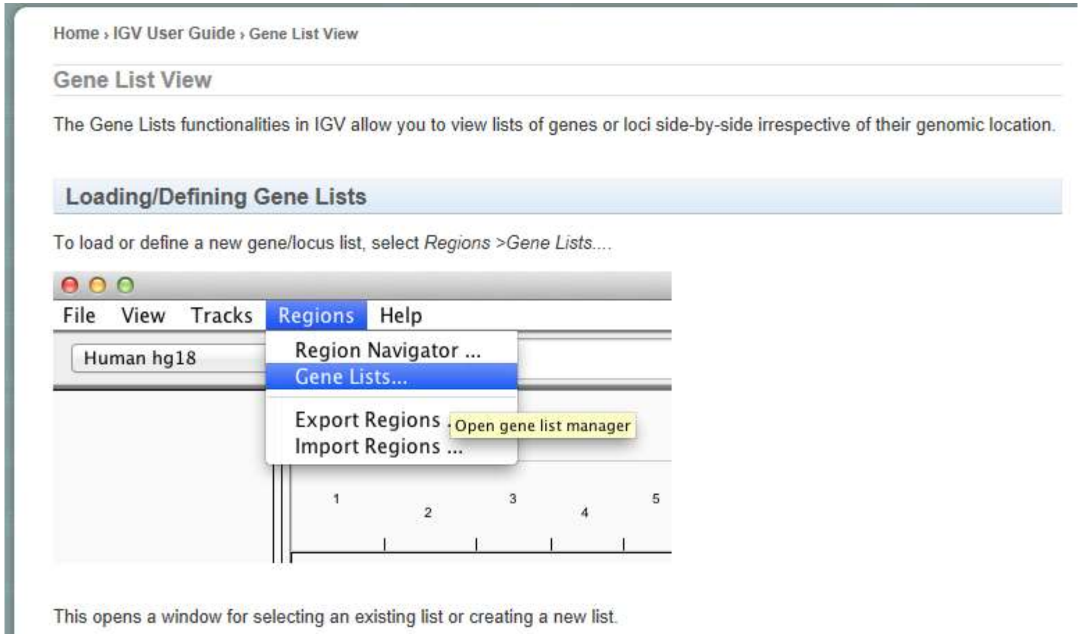
RNA-Seqのデータ表示させる

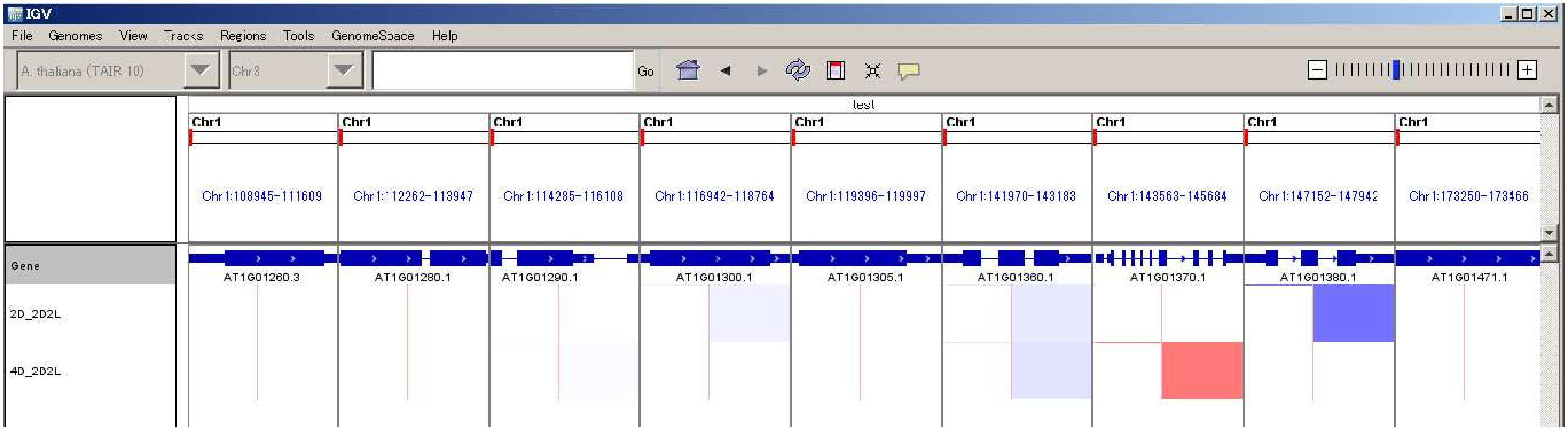
GCTファイルでgene ローカスの発現情報を図示

```
#
#
Name        Description             2D_2D2L     4D_2D2L
ANAC001     |@Chr1:3630-5899        -2.60184    -2.60956
DCL1        |@Chr1:23145-33153      -0.742675   -1.5642
MIR838A     |@Chr1:23145-33153      0           0
AT1G01073   |@Chr1:44676-44787      0           0
IQD18       |@Chr1:52238-54692      -1.93871    -1.13128
AT1G01115   |@Chr1:56623-56740      0           0
GIF2        |@Chr1:72338-74737      -0.251287   -0.616679
AT1G01180   |@Chr1:75582-76758      0.45929     -0.809567
AT1G01210   |@Chr1:88897-89745      1.6964      0.857196
FKGP        |@Chr1:91375-95651      -0.174589   0.725947
AT1G01240   |@Chr1:99893-101834     -0.226384   -0.936641
AT1G01260   |@Chr1:108945-111609    -0.161848   0.315699
CYP703A2    |@Chr1:112262-113947    0           0
CNX3        |@Chr1:114285-116108    0.111249    -0.551359
AT1G01300   |@Chr1:116942-118764    -0.68348    0.108578
```



Gene listを定義して
サンプルごと
条件ごと
の発現・発現変動を
カラーマップできる





IGV実習

IGV Desktop Application

Search

- About IGV
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  - Other versions of IGV
- Release Notes
- Getting Help

Download IGV

IGV version 2.19.1

IGV for Windows
Java included

What's New: See the Release Notes for what's new in each IGV release.

All platforms

IGV requires Java 21 or greater as of IGV 2.19.1. If you download one of the IGV versions that does not include Java, make sure you have Java 21 installed and in your path.

Mac users: As of IGV 2.19.1, the IGV Mac apps require **MacOS 11 (Big Sur)** or greater.

- To run **the latest IGV** on older MacOS versions, download and use the ***Command line IGV for all platforms*** option from the main Download page.
- To run **a working IGV Mac app** on older MacOS versions, download the **2.18.4 Mac app** (includes Java). Or load any IGV Mac app version 2.18.4 or earlier from the IGV download archive.

Linux users: The *IGV for Linux* download includes AdoptOpenJDK (now Eclipse Temurin) version 21 for x64 Linux. See their list of supported platforms here. If your platform is not on the "x64 Linux" list, or the packaged Java does not work on your version of Linux, download the *"Command line IGV for all platforms"* and use it with your own Java installation.

IGV for MacOS
Apple Chip - Java included

IGV for MacOS
Intel Chip - Java included

IGV for MacOS
Separate Java 21 required

IGV for Windows
Java included

IGV for Windows
Separate Java 21 required

IGV for Linux
Java included

Command line IGV and igvtools for all platforms
Separate Java 21 required

Other releases of IGV Desktop

Development snapshot build. Latest development snapshot; built at least nightly

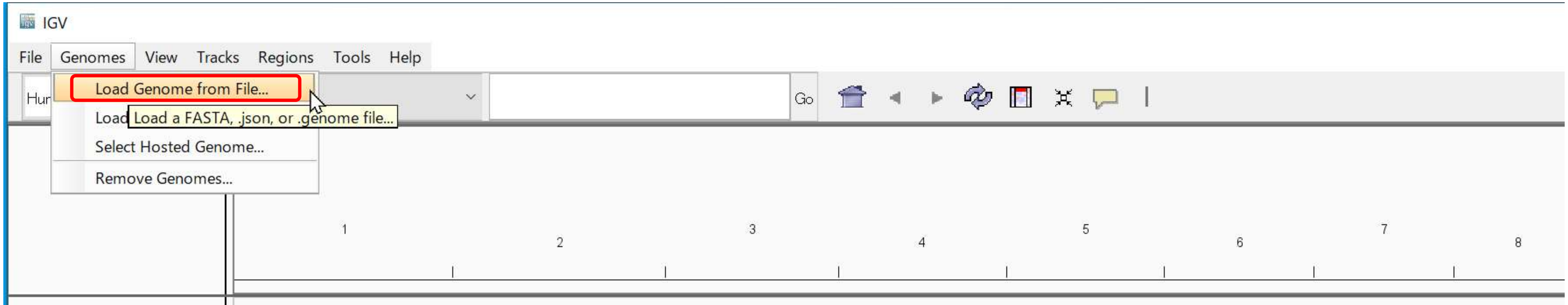
Archived releases. Old releases going back to IGV 2.0

IGVの使用法を学ぶと共に
先のファイルフォーマットも
確認しよう

以下のファイルを確認

buc.genome.fasta
buc.gtf
buc_cg.wig
illumina_ex_B2_Read_bowtie2.mate.sort.bam
illumina_ex_B2_Read_bowtie2.mate.sort.bam.bai
illumina_ex_B4_Read_bowtie2.mate.sort.bam
illumina_ex_B4_Read_bowtie2.mate.sort.bam.bai

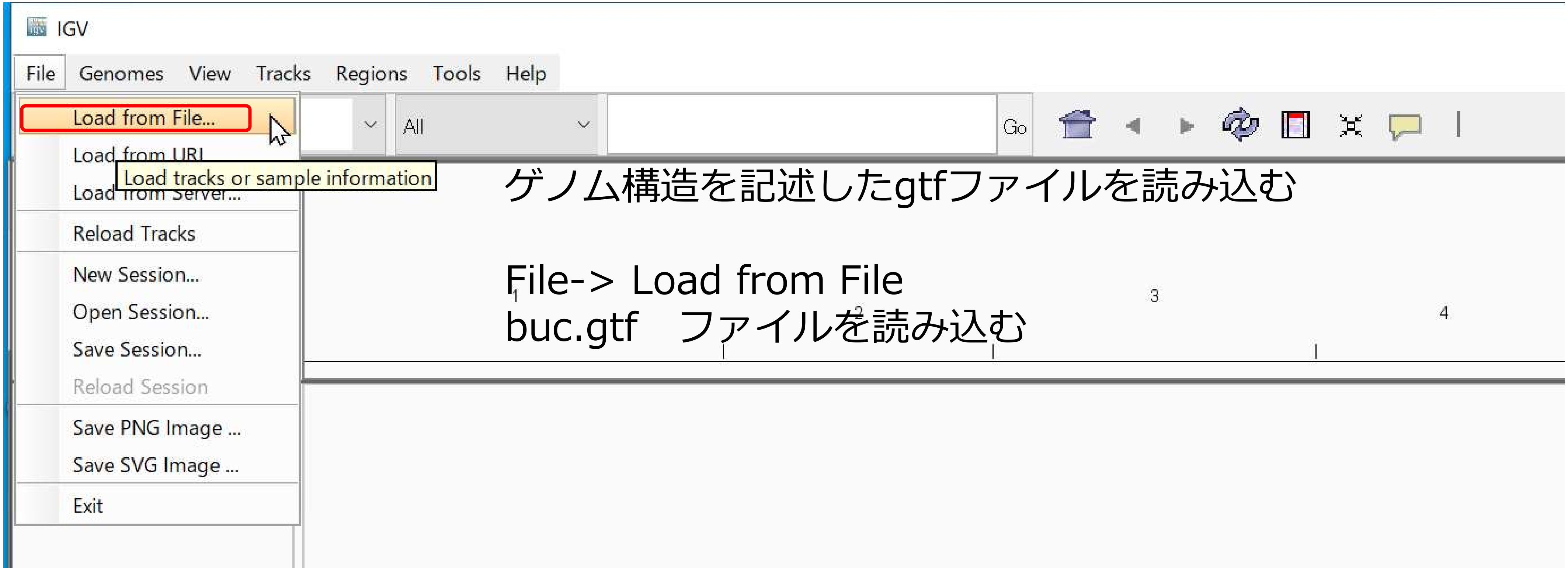
共生バクテリア 2系統を比較する



登録されていない生物種・配列でも、自分でimportすればOK

Buchneraゲノムを使う

Genome -> Load Genome from File
buc.genome.fasta
を読み込む

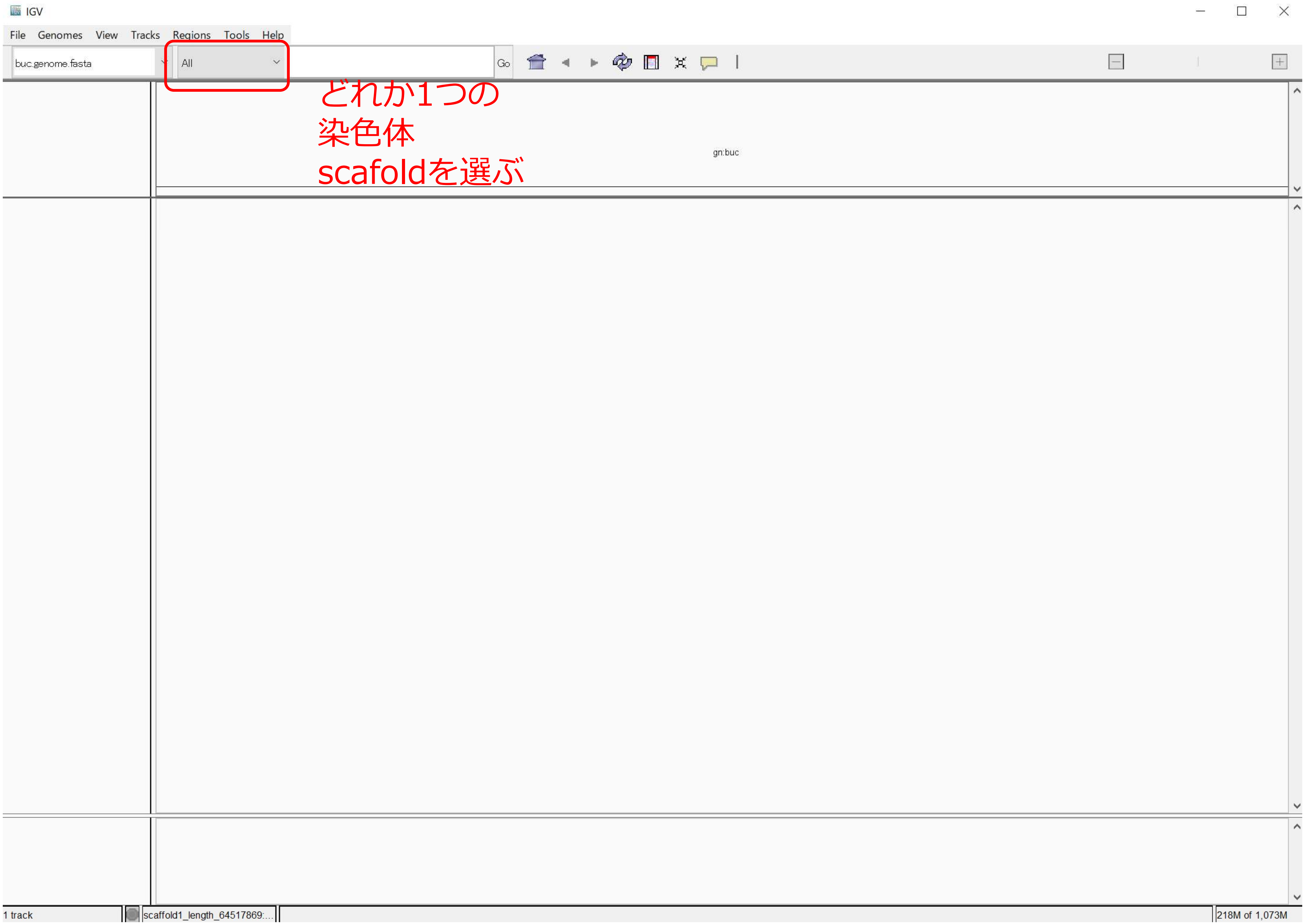


ゲノム構造を記述したgtfファイルを読み込む

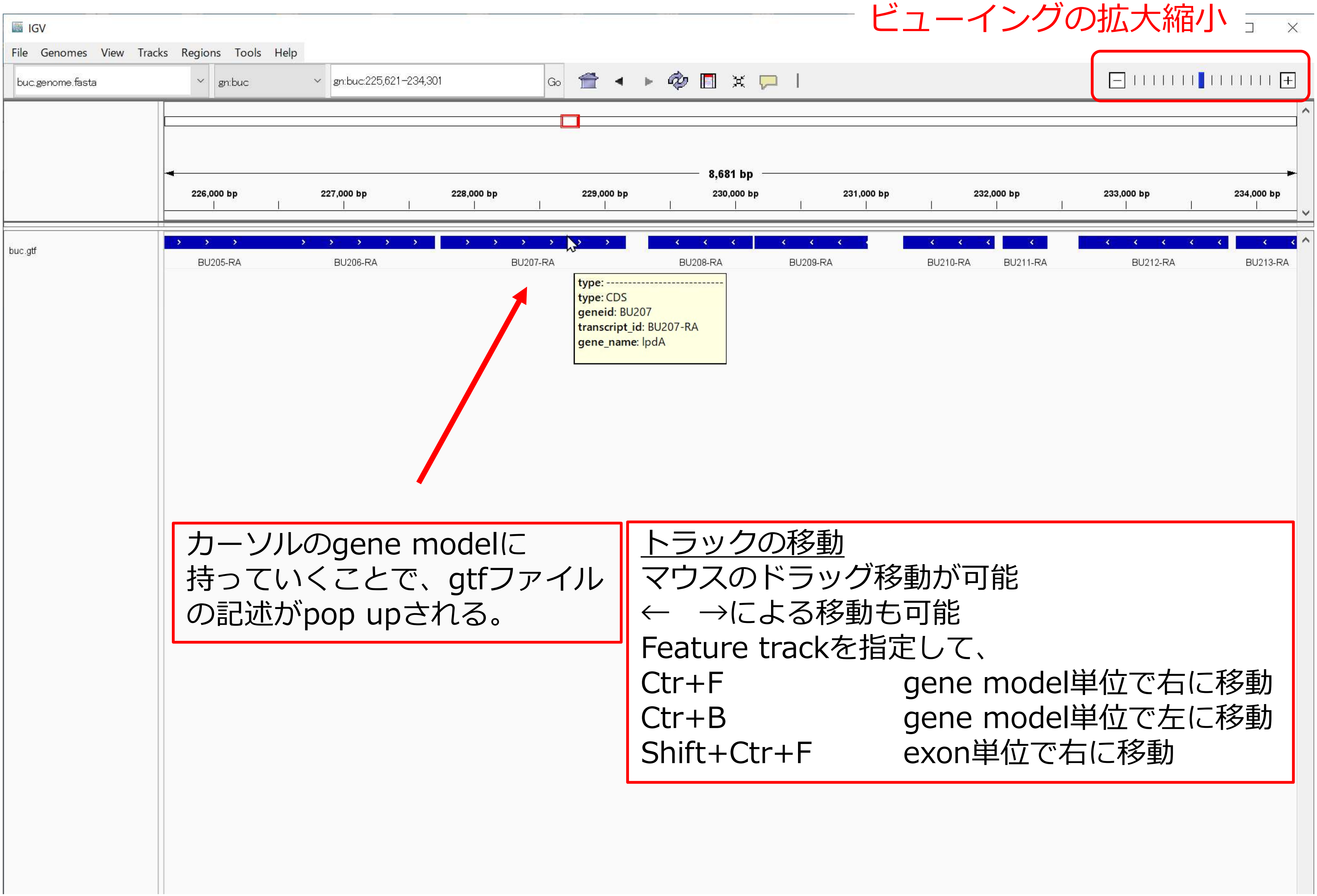
File-> Load from File
buc.gtf　ファイルを読み込む

buc.gtf

gn:buc	KEGG:GENES	CDS	197	2083	.+	geneid "BU001"; transcript_id "BU001-RA"; gene_name "gidA";
gn:buc	KEGG:GENES	CDS	2278	3102	.+	geneid "BU002"; transcript_id "BU002-RA"; gene_name "atpB";
gn:buc	KEGG:GENES	CDS	3139	3378	.+	geneid "BU003"; transcript_id "BU003-RA"; gene_name "atpE";
gn:buc	KEGG:GENES	CDS	3497	3982	.+	geneid "BU004"; transcript_id "BU004-RA"; gene_name "atpF";
gn:buc	KEGG:GENES	CDS	3982	4515	.+	geneid "BU005"; transcript_id "BU005-RA"; gene_name "atpH";
gn:buc	KEGG:GENES	CDS	4530	6068	.+	geneid "BU006"; transcript_id "BU006-RA"; gene_name "atpA";
gn:buc	KEGG:GENES	CDS	6101	6973	.+	geneid "BU007"; transcript_id "BU007-RA"; gene_name "atpG";
gn:buc	KEGG:GENES	CDS	6997	8394	.+	geneid "BU008"; transcript_id "BU008-RA"; gene_name "atpD";
gn:buc	KEGG:GENES	CDS	8421	8837	.+	geneid "BU009"; transcript_id "BU009-RA"; gene_name "atpC";
gn:buc	KEGG:GENES	CDS	8911	11322	.-	geneid "BU010"; transcript_id "BU010-RA"; gene_name "gyrB";
gn:buc	KEGG:GENES	CDS	11449	12549	.-	geneid "BU011"; transcript_id "BU011-RA"; gene_name "dnaN";
gn:buc	KEGG:GENES	CDS	12554	13918	.-	geneid "BU012"; transcript_id "BU012-RA"; gene_name "dnaA";
gn:buc	KEGG:GENES	CDS	14369	14512	.+	geneid "BU013"; transcript_id "BU013-RA"; gene_name "rpmH";
gn:buc	KEGG:GENES	CDS	14525	14872	.+	geneid "BU014"; transcript_id "BU014-RA"; gene_name "rnpA";
gn:buc	KEGG:GENES	CDS	15011	16609	.+	geneid "BU015"; transcript_id "BU015-RA"; gene_name "yidC";


IGV
File Genomes View Tracks Regions Tools Help
buc.genome.fasta
All
Go
どれか1つの
染色体
scafoldを選ぶ
gn:buc
1 track
scaffold1_length_64517869:...
218M of 1,073M

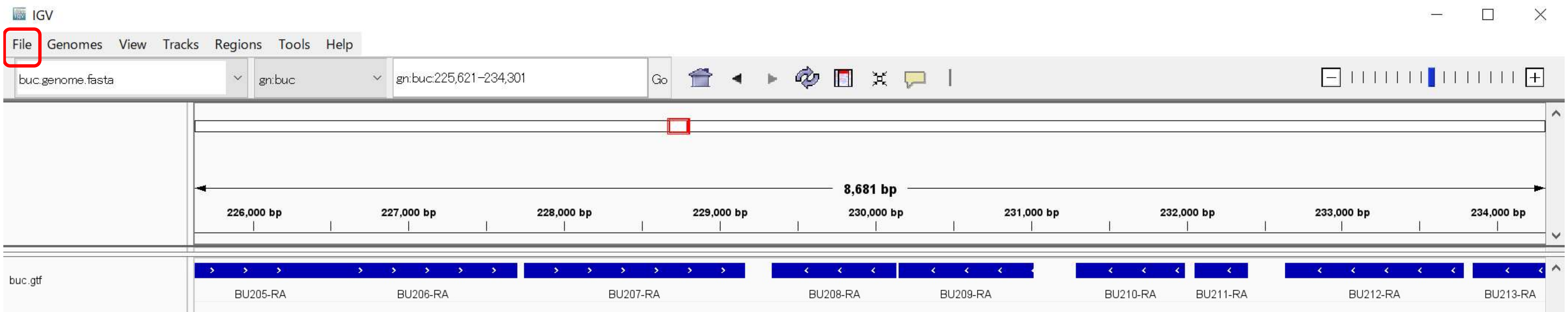
ビューイングの拡大縮小



カーソルのgene modelに持っていくことで、gtfファイルの記述がpop upされる。

<u>トラックの移動</u>
マウスのドラッグ移動が可能
← →による移動も可能
Feature trackを指定して、

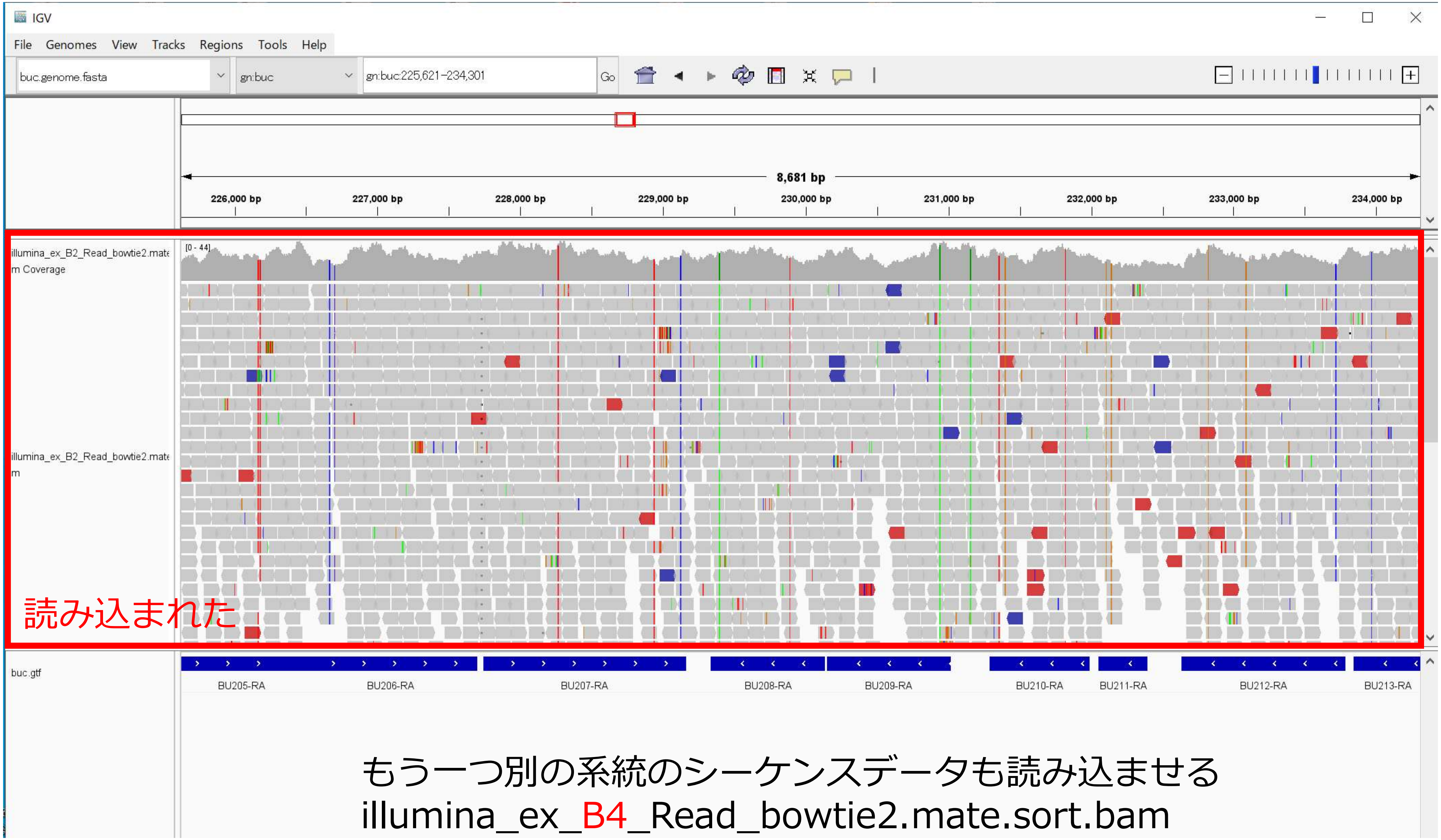
Ctr+F	gene model単位で右に移動
Ctr+B	gene model単位で左に移動
Shift+Ctr+F	exon単位で右に移動



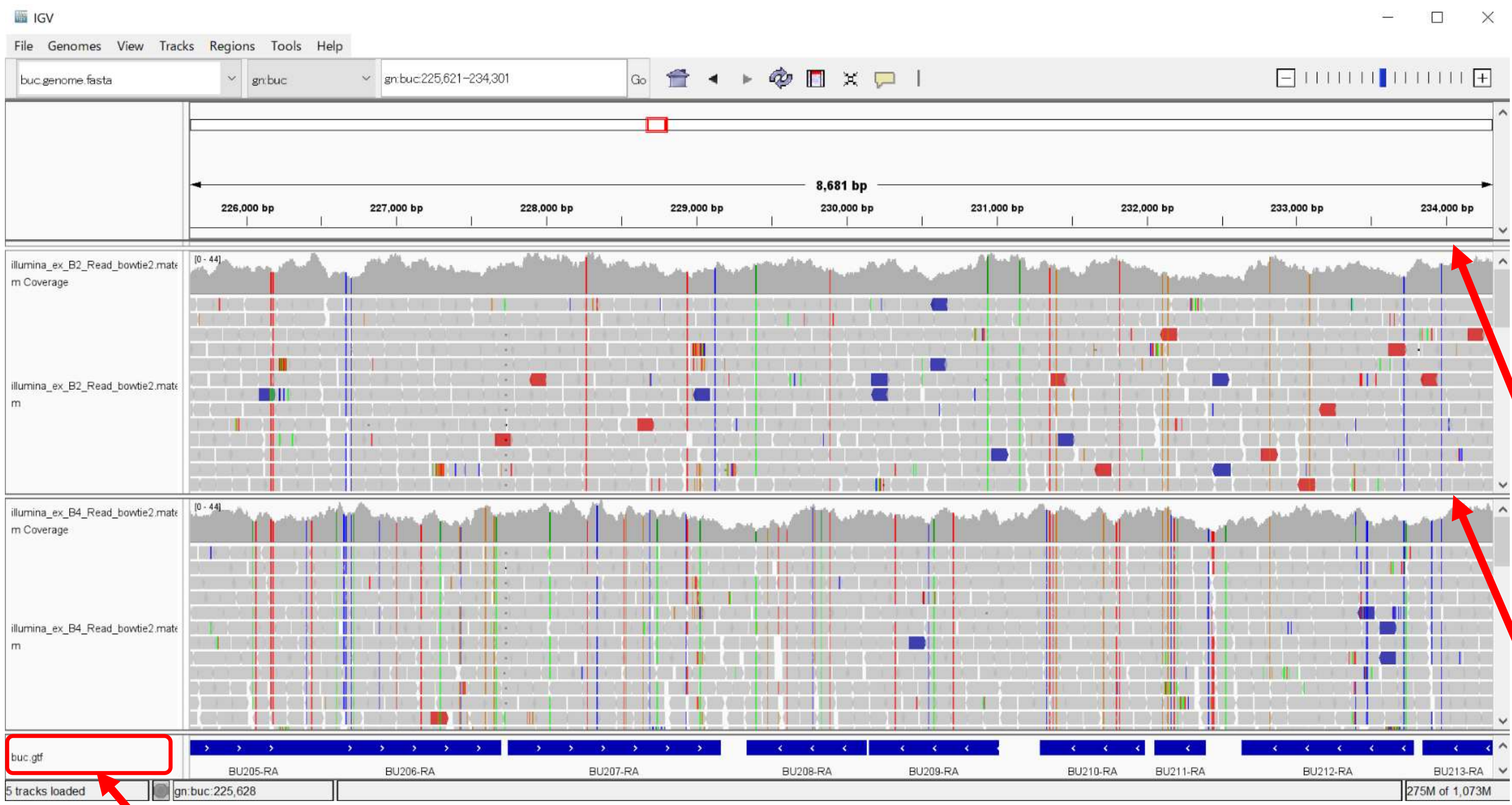
シーケンスデータを読み込ませる
File -> Loead from File

illumina_ex_B2_Read_bowtie2.mate.sort.bam

同ファイル名のindexファイル
illumina_ex_B2_Read_bowtie2.mate.sort.bam.bai
が同じフォルダーにないといけない。

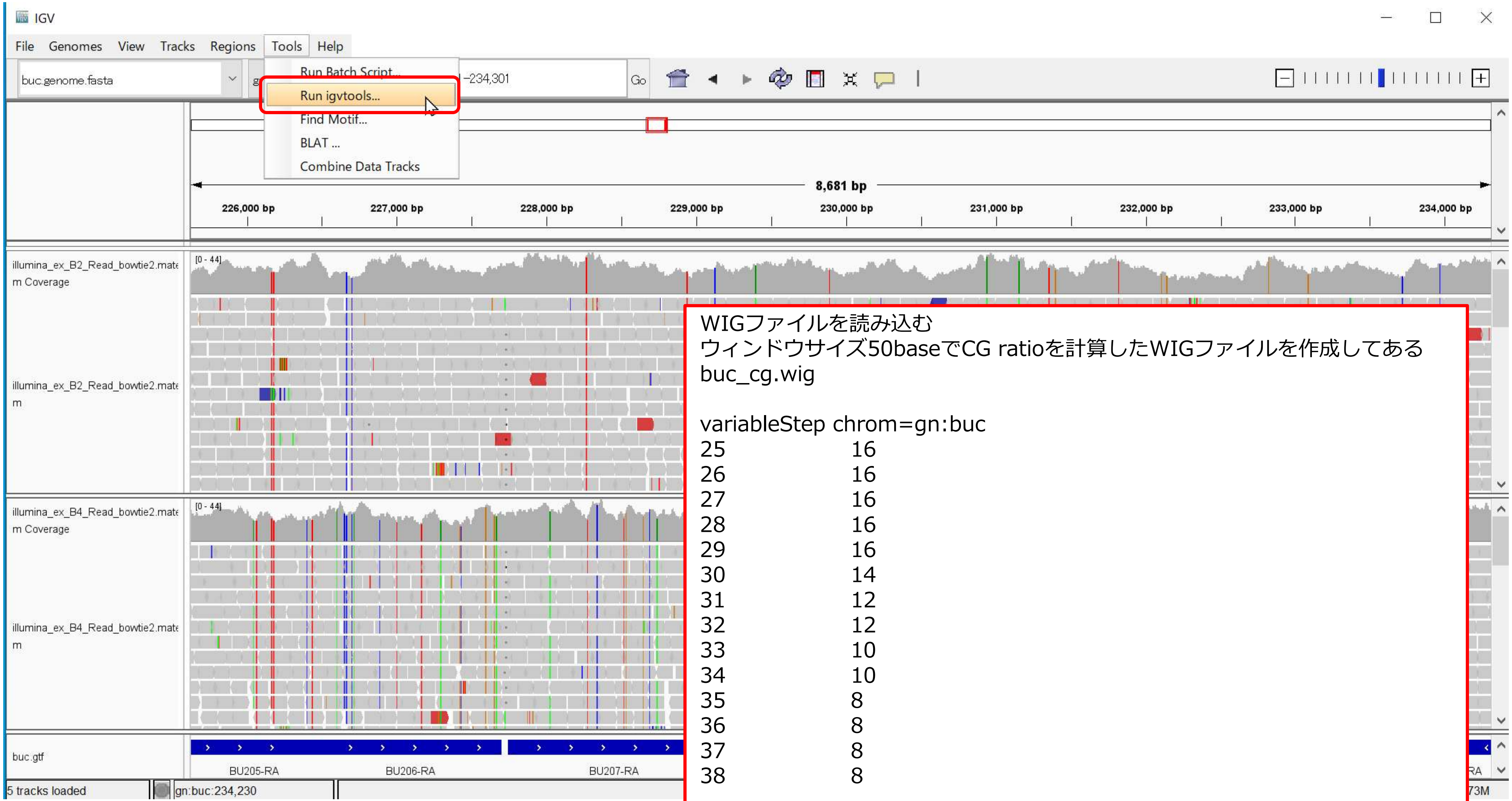


もう一つ別の系統のシーケンスデータも読み込ませる
illumina_ex_B4_Read_bowtie2.mate.sort.bam



各Trackはドラッグアンドドロップで移動できる

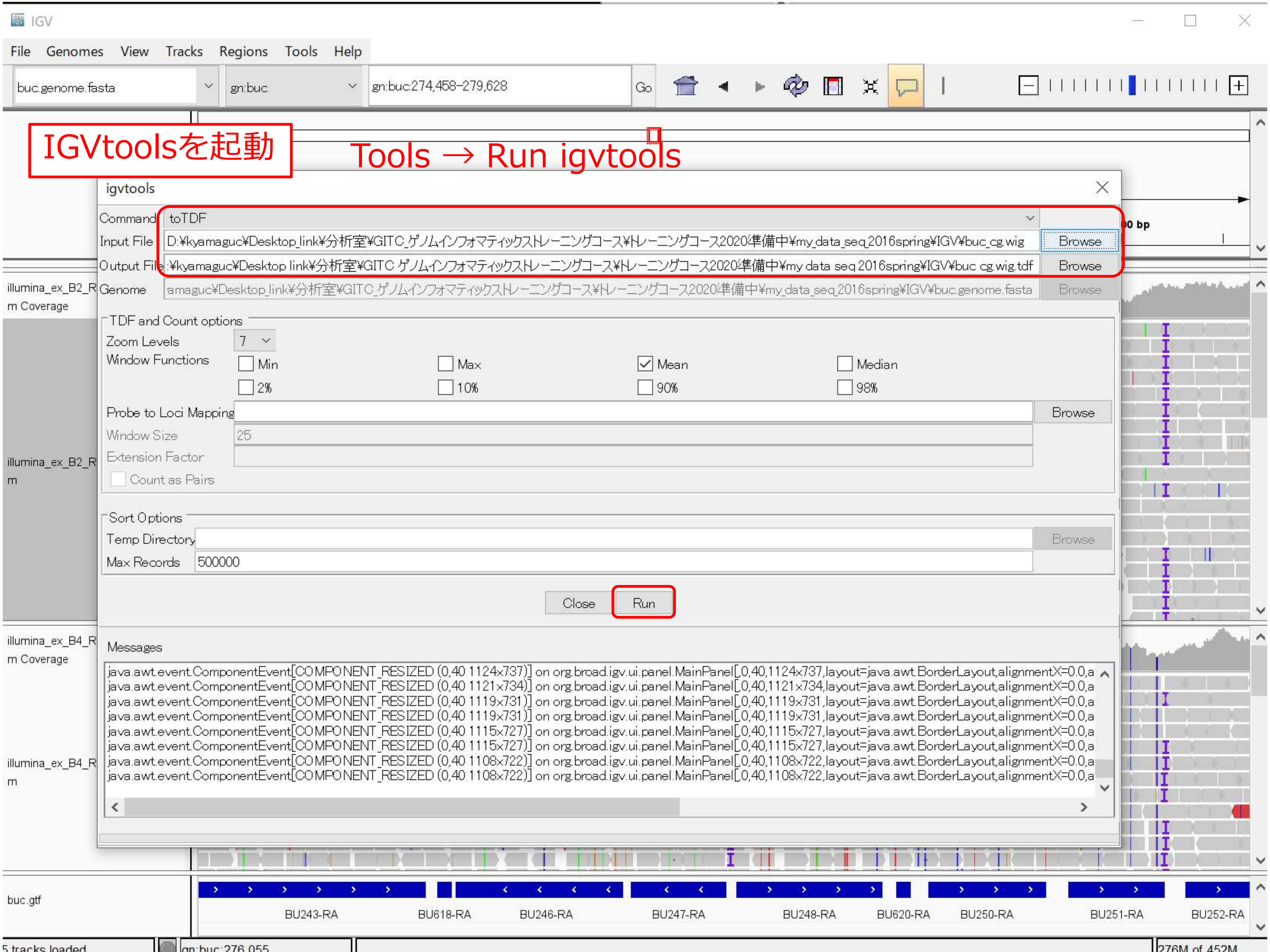
こういう所を
クリックしたまま
移動させることで
幅を変更



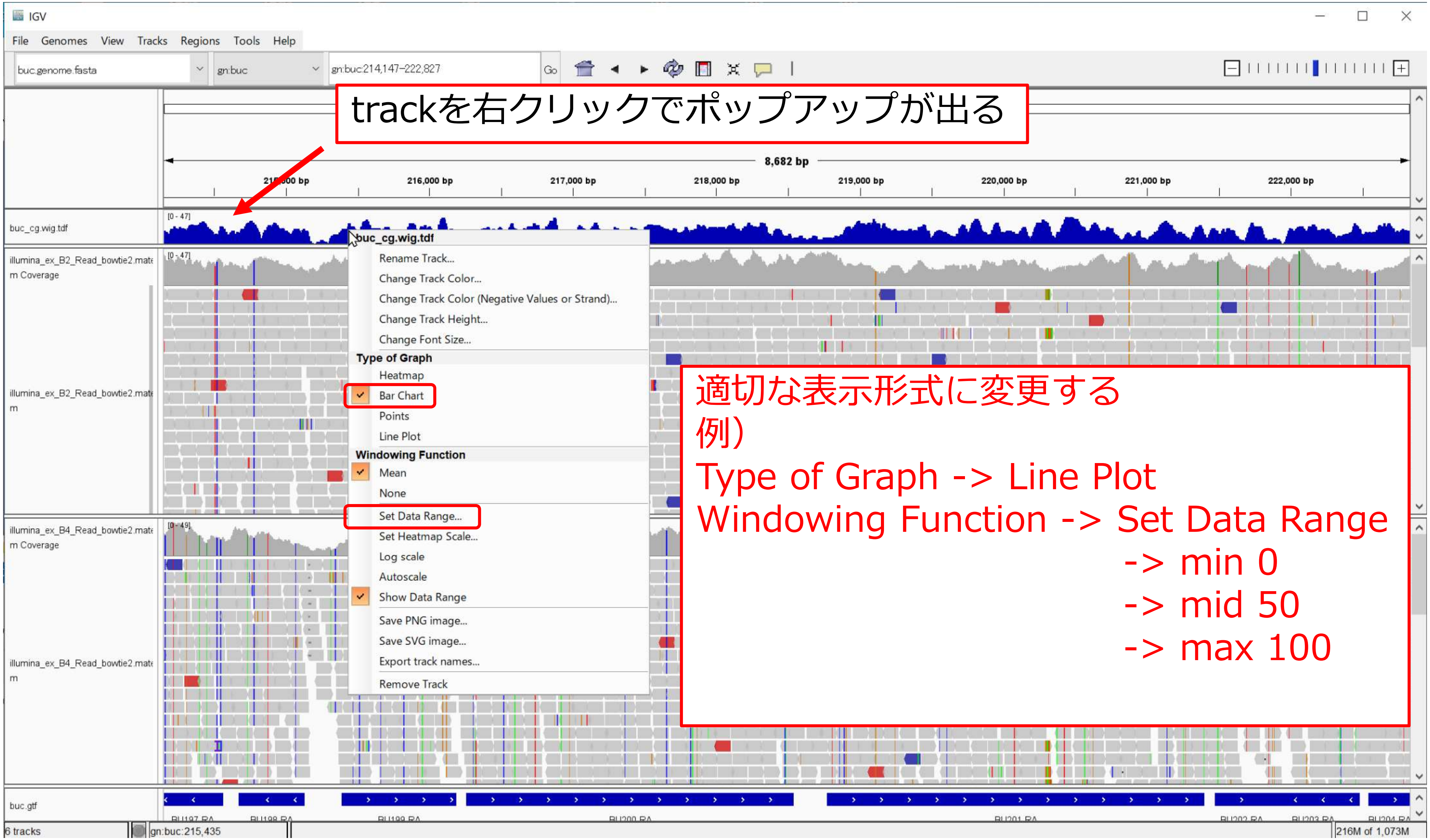
WIGファイルを読み込む
ウィンドウサイズ50baseでCG ratioを計算したWIGファイルを作成してある
buc_cg.wig

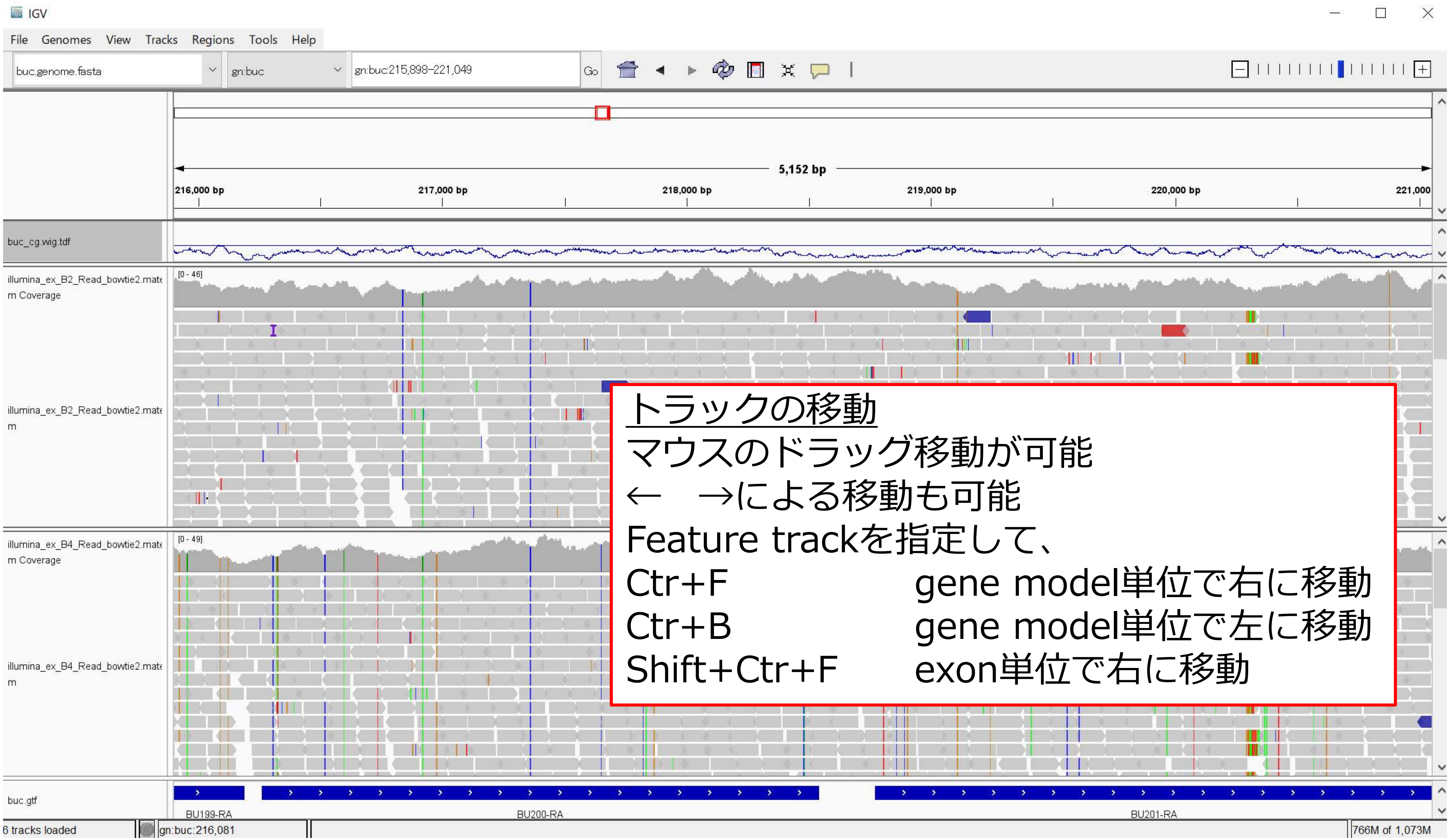
```
variableStep chrom=gn:buc
25      16
26      16
27      16
28      16
29      16
30      14
31      12
32      12
33      10
34      10
35      8
36      8
37      8
38      8
```

File -> Load from File でそのまま読み込ませることも可能だが、
テキストファイルで膨大なサイズだと非常に時間がかかる。
Tools -> Run igvtools


IGVtoolsを起動
Tools → Run igvtools
igvtools
Command toTDF
Input File D:¥kyamaguc¥Desktop_link¥分析室¥GITC_ゲノムインフォマティックストレーニングコース¥トレーニングコース2020準備中¥my_data_seq_2016spring¥IGV¥buc_cg.wig
Output File ¥kyamaguc¥Desktop_link¥分析室¥GITC_ゲノムインフォマティックストレーニングコース¥トレーニングコース2020準備中¥my_data_seq_2016spring¥IGV¥buc_cg.wig.tdf
Genome amaguc¥Desktop_link¥分析室¥GITC_ゲノムインフォマティックストレーニングコース¥トレーニングコース2020準備中¥my_data_seq_2016spring¥IGV¥buc.genome.fasta
Browse
TDF and Count options
Zoom Levels 7
Window Functions
Min
Max
Mean
Median
2%
10%
90%
98%
Probe to Loci Mapping
Window Size 25
Extension Factor
Count as Pairs
Sort Options
Temp Directory
Max Records 500000
Close
Run
Messages
File Genomes View Tracks Regions Tools Help
buc.genome.fasta
gn:buc
gn:buc:274,458-279,628
Go
buc.gtf
BU243-RA
BU618-RA
BU246-RA
BU247-RA
BU248-RA
BU620-RA
BU250-RA
BU251-RA
BU252-RA

WIGファイルがバイナリー化されたtdfファイルができるので、
それをFile -> Load from file


IGV
File Genomes View Tracks Regions Tools Help
trackを右クリックでポップアップが出る
buc_cg.wig.tdf
Rename Track...
Change Track Color...
Change Track Color (Negative Values or Strand)...
Change Track Height...
Change Font Size...
Type of Graph
Heatmap
Bar Chart
Points
Line Plot
Windowing Function
Mean
None
Set Data Range...
Set Heatmap Scale...
Log scale
Autoscale
Show Data Range
Save PNG image...
Save SVG image...
Export track names...
Remove Track
適切な表示形式に変更する
例）
Type of Graph -> Line Plot
Windowing Function -> Set Data Range
-> min 0
-> mid 50
-> max 100



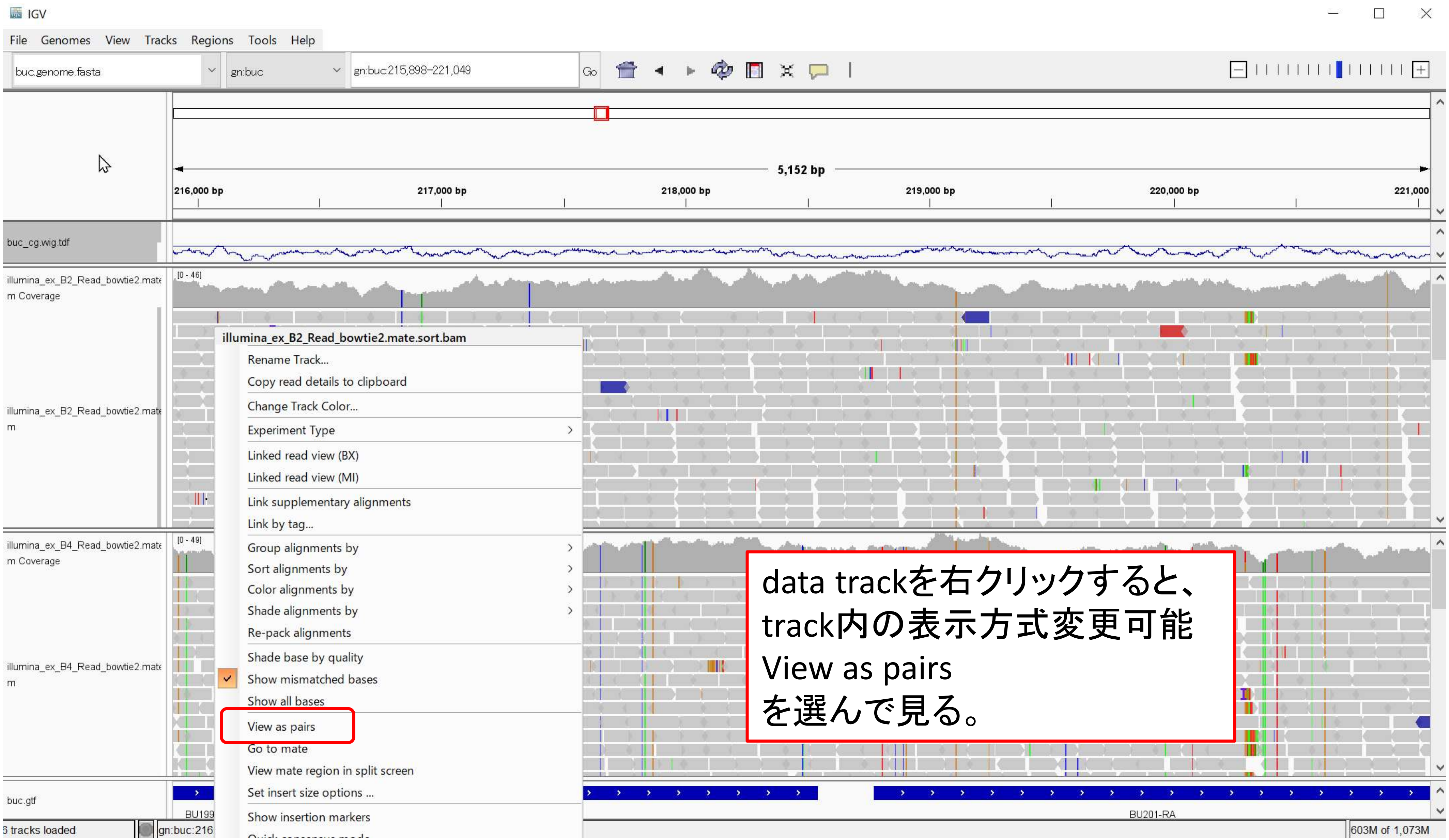
<u>トラックの移動</u>

マウスのドラッグ移動が可能

← →による移動も可能

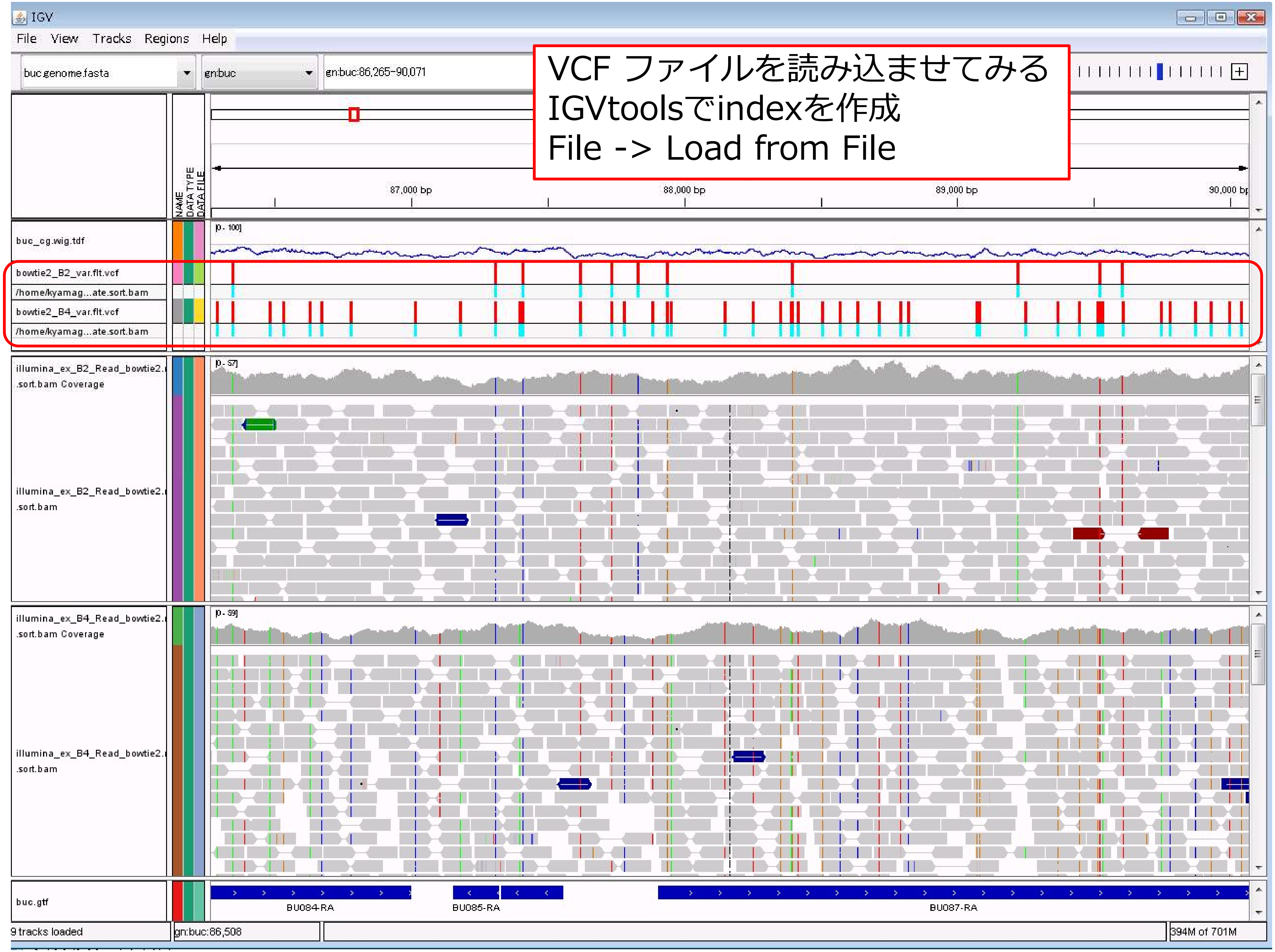
Feature trackを指定して、

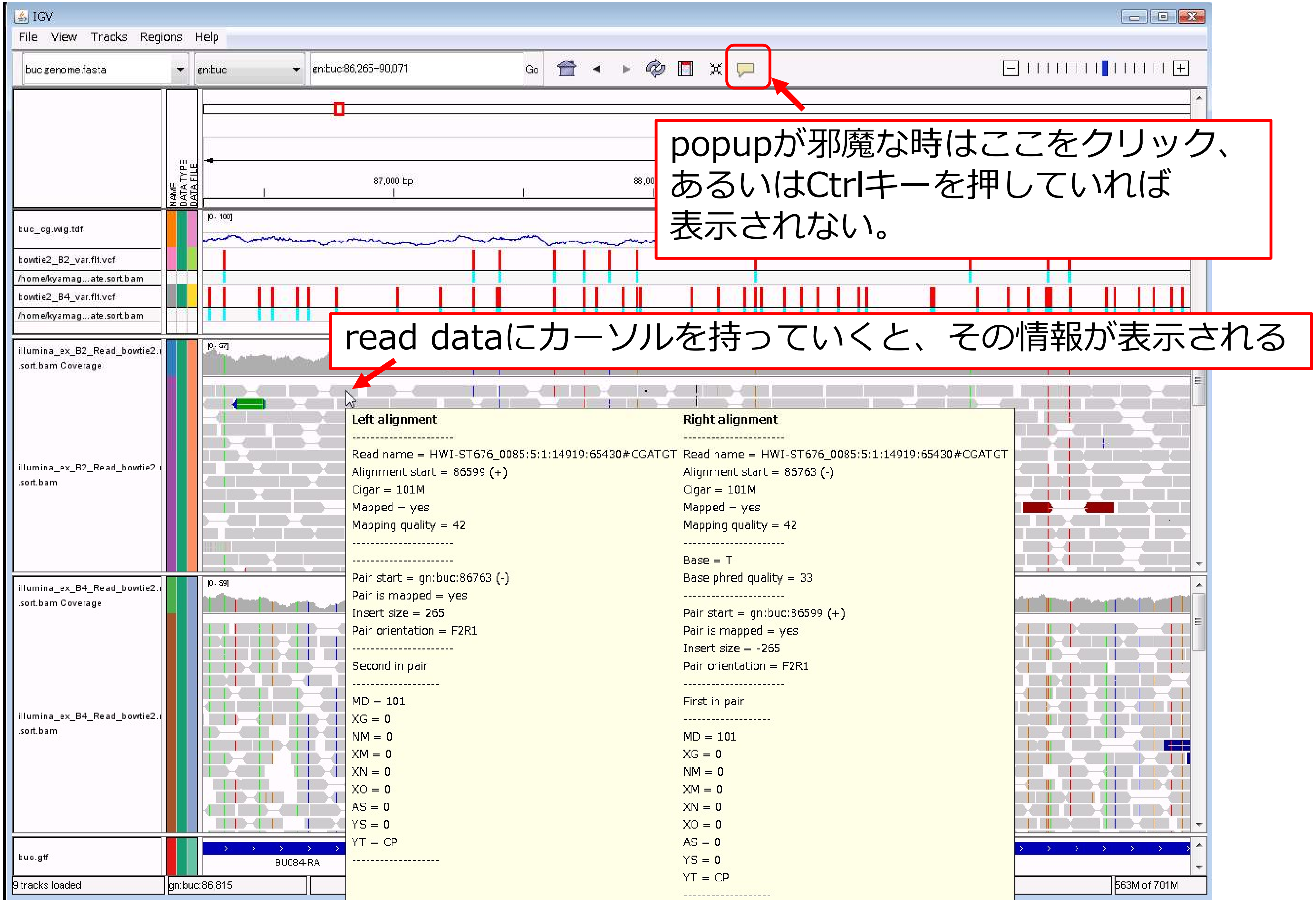
Ctr+F	gene model単位で右に移動
Ctr+B	gene model単位で左に移動
Shift+Ctr+F	exon単位で右に移動


IGV
File Genomes View Tracks Regions Tools Help
buc.genome.fasta
gn:buc
gn:buc:215,898-221,049
Go
5,152 bp
216,000 bp
217,000 bp
218,000 bp
219,000 bp
220,000 bp
221,000
buc_cg.wig.tdf
illumina_ex_B2_Read_bowtie2.mate m Coverage
[0 - 46]
illumina_ex_B2_Read_bowtie2.mate m
illumina_ex_B4_Read_bowtie2.mate m Coverage
[0 - 49]
illumina_ex_B4_Read_bowtie2.mate m
buc.gtf
BU201-RA
6 tracks loaded
603M of 1,073M
illumina_ex_B2_Read_bowtie2.mate.sort.bam
Rename Track...
Copy read details to clipboard
Change Track Color...
Experiment Type
Linked read view (BX)
Linked read view (MI)
Link supplementary alignments
Link by tag...
Group alignments by
Sort alignments by
Color alignments by
Shade alignments by
Re-pack alignments
Shade base by quality
Show mismatched bases
Show all bases
View as pairs
Go to mate
View mate region in split screen
Set insert size options ...
Show insertion markers
data trackを右クリックすると、
track内の表示方式変更可能
View as pairs
を選んで見る。


IGV
File View Tracks Regions Help
buc genome.fasta
gn:buc
gn:buc:316,735-323,248
Go
317,000 bp
23,000 bp
pair endを維持した表示になる
insertion, deletionを見たい場合はこちらが良い。
buc_cg.wig.tdf
illumina_ex_B2_Read_bowtie2.
.sort.bam Coverage
illumina_ex_B2_Read_bowtie2.
.sort.bam
illumina_ex_B4_Read_bowtie2.
.sort.bam Coverage
illumina_ex_B4_Read_bowtie2.
.sort.bam
buc.gtf
BU289-RA
BU290-RA
BU291-RA
BU292-RA
BU293-RA
BU294-RA
7 tracks
gn:buc:318,057
254M of 493M

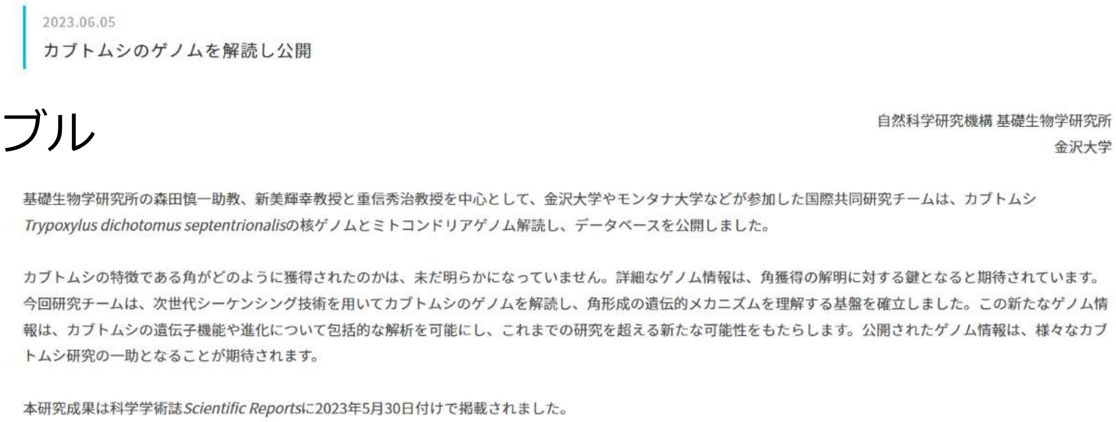
VCF ファイルを読み込ませてみる
IGVtoolsでindexを作成
File -> Load from File





IGV
File View Tracks Regions Help
buc genome.fasta
gn:buc
gn:buc:86,265-90,071
Go
popupが邪魔な時はここをクリック、
あるいはCtrlキーを押していれば
表示されない。
87,000 bp
NAME
DATA TYPE
DATA FILE
buc_cg.wig.tdf
bowtie2_B2_var.flt.vcf
/home/kyamag...ate.sort.bam
bowtie2_B4_var.flt.vcf
/home/kyamag...ate.sort.bam
read dataにカーソルを持っていくと、その情報が表示される
illumina_ex_B2_Read_bowtie2.sort.bam Coverage
illumina_ex_B2_Read_bowtie2.sort.bam
illumina_ex_B4_Read_bowtie2.sort.bam Coverage
illumina_ex_B4_Read_bowtie2.sort.bam
buc.gtf
BU084-RA
9 tracks loaded
gn:buc:86,815
563M of 701M
Left alignment
Read name = HWI-ST676_0085:5:1:14919:65430#CGATGT
Alignment start = 86599 (+)
Cigar = 101M
Mapped = yes
Mapping quality = 42
Pair start = gn:buc:86763 (-)
Pair is mapped = yes
Insert size = 265
Pair orientation = F2R1
Second in pair
MD = 101
XG = 0
NM = 0
XM = 0
XN = 0
XO = 0
AS = 0
YS = 0
YT = CP
Right alignment
Read name = HWI-ST676_0085:5:1:14919:65430#CGATGT
Alignment start = 86763 (-)
Cigar = 101M
Mapped = yes
Mapping quality = 42
Base = T
Base phred quality = 33
Pair start = gn:buc:86599 (+)
Pair is mapped = yes
Insert size = -265
Pair orientation = F2R1
First in pair
MD = 101
XG = 0
NM = 0
XM = 0
XN = 0
XO = 0
AS = 0
YS = 0
YT = CP

Gene prediction結果をIGVで見ながら考察する


2023.06.05
カブトムシのゲノムを解読し公開

自然科学研究機構 基礎生物学研究所
金沢大学

基礎生物学研究所の森田慎一助教、新美輝幸教授と重信秀治教授を中心として、金沢大学やモンタナ大学などが参加した国際共同研究チームは、カブトムシ *Trypoxylus dichotomus septentrionalis*の核ゲノムとミトコンドリアゲノム解読し、データベースを公開しました。

カブトムシの特徴である角がどのように獲得されたのかは、未だ明らかになっていません。詳細なゲノム情報は、角獲得の解明に対する鍵となると期待されています。今回研究チームは、次世代シーケンシング技術を用いてカブトムシのゲノムを解読し、角形成の遺伝的メカニズムを理解する基盤を確立しました。この新たなゲノム情報は、カブトムシの遺伝子機能や進化について包括的な解析を可能にし、これまでの研究を超える新たな可能性をもたらします。公開されたゲノム情報は、様々なカブトムシ研究の一助となることが期待されます。

本研究成果は科学学術誌*Scientific Reports*に2023年5月30日付けで掲載されました。

Linked read sequenceをsupernovaでアセンブル
Scaffold N50 8.02Mbのアセンブルを得た
実習用にscaffold27(15,574,170bp)のみ
取り出した
fastaファイル　　kabuto_part27.fasta

論文で報告している、short readによるRNA-Seq
と近縁種のタンパク質配列をベースにgene predictionした、
23,987 protein-coding sequenceを記述した
gffファイル　　　p04_part27.gff

実習
これらをIGVを使って可視化してみよう

カブトムシアセンブル結果の以下fastaファイルを読み込む
Genomes -> Load Genome from File
kabuto_part27.fasta

protein-coding sequenceを記述したgffファイルを読み込む
File -> Load from File
p04_part27.gff

全長cDNA配列をハイスループットで得る新技術

全長cDNAをそのままシーケンスするISO-Seq法
さらにcDNAを連結して、ハイスループットにシーケンス

pacbio社公開資料より

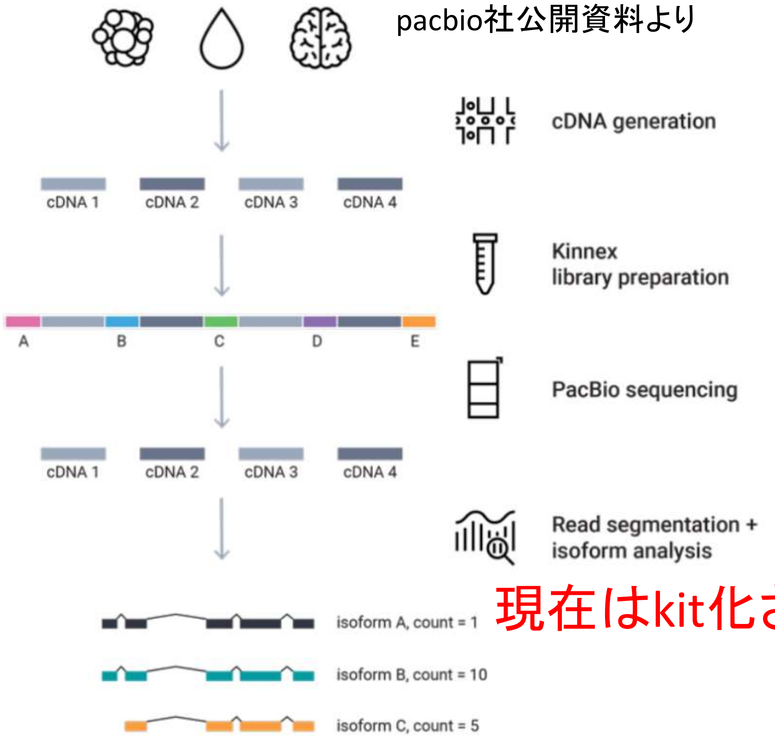


Figure 2. Kinnex full-length RNA sequencing. Full-length cDNA molecules are concatenated into a large-insert library, sequenced, then processed using the PacBio software.

現在はkit化された製品が利用可能

nature biotechnology

Brief Communication https://doi.org/10.1038/s41587-023-01815-7

High-throughput RNA isoform sequencing using programmed cDNA concatenation

Received: 1 October 2021

Accepted: 2 May 2023

Published online: 08 June 2023

Check for updates

Aziz M. Al'Khafaji[1,11], Jonathan T. Smith[1,11], Kiran V. Garimella[1,11], Mehrtash Babadi[1,11], Victoria Popic[1,11], Moshe Sade-Feldman[1,2], Michael Gatzen[1], Siranush Sarkizova[1], Marc A. Schwartz[1,3,4,5], Emily M. Blaum[1,2], Allyson Day[1], Maura Costello[1], Tera Bowers[1], Stacey Gabriel[1], Eric Banks[1], Anthony A. Philippakis[1], Genevieve M. Boland[6], Paul C. Blainey[1,7,8] & Nir Hacohen[1,2,9,10]

Full-length RNA-sequencing methods using long-read technologies can capture complete transcript isoforms, but their throughput is limited. We introduce multiplexed arrays isoform sequencing (MAS-ISO-seq), a technique for programmably concatenating complementary DNAs (cDNAs) into molecules optimal for long-read sequencing, increasing the throughput >15-fold to nearly 40 million cDNA reads per run on the Sequel IIe sequencer. When applied to single-cell RNA sequencing of tumor-infiltrating T cells, MAS-ISO-seq demonstrated a 12- to 32-fold increase in the discovery of differentially spliced genes.

short readからのアセンブルで生じるミスアセンブルを排除
オルタナティブスプライシング状況が明確に分かる
隣接する遺伝子を明確に区分け

Kinnex full-length RNA kit

この結果をゲノムにmappingした
bamファイル　　mapped_part27.bam

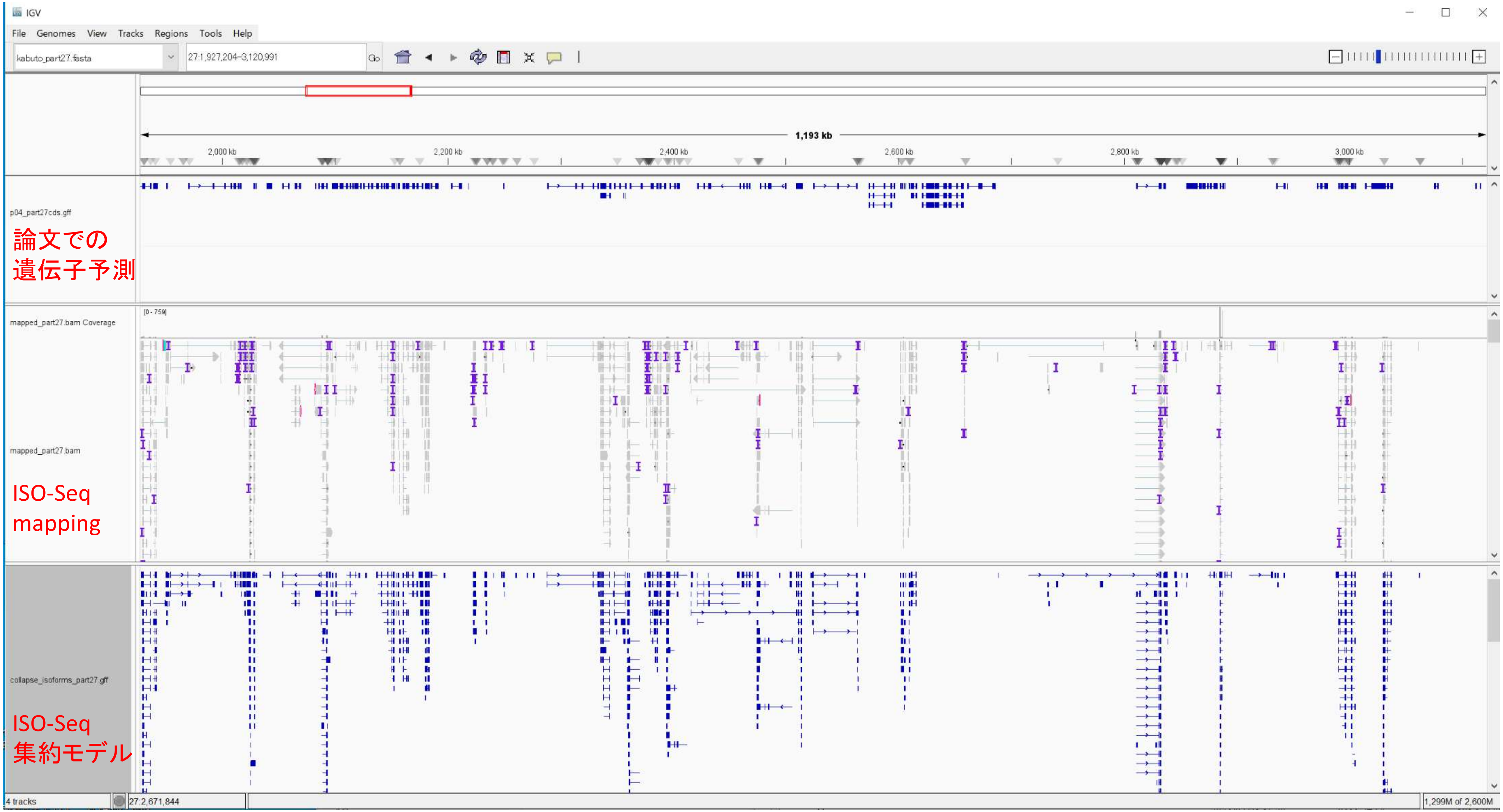
同一配列を集約した
gffファイル　　collapse_isoforms_part27.gff

を用意した

実習
これら2つのファイルを追加しよう

File -> Loead from File
mapped_part27.bam

File -> Loead from File
collapse_isoforms_part27.gff



可視化によって直感的な比較検討が可能に。

IGV紹介のまとめ

可視化ツールとして十分な機能を持つ

・無料

・比較的簡単・お手軽

・自分で見るためにも良し、人に見せるためにも良し

・利用範囲は次世代DNAシーケンサーに限定しない
広くゲノミクスの解析に有用

ごく一部のみの機能を紹介しました。
ウェブサイトを見ながら復習をお勧めします。